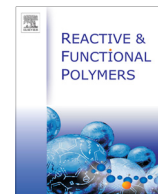




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Review

Targeted gas separations through polymer membrane functionalization

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ABSTRACT

Gas separation membranes have been studied for several decades and are starting to find commercial acceptance. This review will focus on polymer functionalization to improve gas separation performance, namely, permeability, permselectivity, or both. The review will cover both “diffusivity controlled” and “solubility controlled” functionalization strategies; each strategy refers to the different mode of gas transport through the membrane. Diffusivity controlled functionalization strategies mainly involve control over free volume elements in amorphous polymers via promotion or inhibition of chain packing through functional groups. As such, the effects of this functionalization are typically confined to the well-known selectivity/permeability tradeoff. Solubility controlled modification strategies utilize functional groups that have strong chemical interactions with some of the penetrant molecules and offers an enhanced solution-diffusion or a non-solution-diffusion permeation pathway. This functionalization can potentially exceed the Robeson upper bound, but is often challenged by impurities and deactivation of the chemical functionality.

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1. Introduction

A combination of the increased prevalence of natural gas, rising CO₂ emissions, and a focus on environmentally-responsible industrial growth has created a growing demand for new, sustainable, energy-efficient technologies. Gas separation membranes are receiving increased attention due to the immense amount of energy and concomitant CO₂ emissions associated with current industrial gas separation technologies. In fact, these status-quo, thermal-based separations consume approximately 5–10% of the United States' primary energy supply [1]. One of the main advantages of gas separation membranes relative to current thermal separations lies in the membrane's ability to perform the separation near the thermodynamic limit of demixing. Another key advantage of membrane technology is the ability to package functional surface area into compact devices, which minimizes the chemical potential driving force necessary to provide the desired gas productivity [2]. The low cost of processing polymers into membranes and assembling them into high surface area to volume modules has made polymeric membranes a promising candidate for industrial gas separation applications.

Polymeric gas separation membranes have already found some use in industrial processes, primarily for small nitrogen generation requirements, hydrogen purification, and offshore natural gas purification. A handful of polymers have been used commercially including: cellulose acetate, polysulfone, polyimides, polyaramides, tetrabromo polycarbonate, polydimethylsiloxane, and polyphenylene oxide [3]. While effective, many of the commercially deployed membranes suffer from low selectivities, low flux, and other issues such as plasticization. There is a significant research effort underway to design and develop new polymeric membranes that can push the limits of productivity and efficiency while maintaining stability in aggressive feed conditions.

This article provides an overview of the state-of-the-art functionalized polymeric membranes for gas separations. We broadly define functionalized polymers as any base polymer with additional chemical groups that impart a change in either the physical or chemical characteristics, with an emphasis on improving the performance for a targeted gas separation application. Functionalization of polymers overcomes some of the inherent limitations of the base polymer while opening up the possibility to combine different desired properties to engineer high performance materials. We have split our definition of functionalization into two groups

that seek to manipulate gas permeation by contrasting means: (i) “diffusivity controlled” functionalization that mainly seeks to control chain packing in glassy polymers and (ii) “solubility controlled” functionalization that seeks to enhance the desired gas solubility. We limit our discussion to the scheme shown in Fig. 1 for glassy polymers while excluding other more exotic classes of gas separation membranes fabricated from stimuli responsive, conducting, interpenetrating, and blended polymers. Finally, we will provide some perspectives and future directions for gas separation membranes.

2. Background

Membranes utilized for gas separations differ from other ultrafiltration (UF), nanofiltration (NF), and reverse osmosis (RO) membranes for a variety of reasons. In these latter separations, the species being separated have size differences on the order of nanometers or larger. Moreover, the fast species in UF, NF, and RO are often substantially smaller—more than three times—than the rejected or slow species. Gas separations are much more difficult to achieve due to less pronounced size differences between penetrating molecules—several tenths of an Ångström difference are common. Table 1 lists the kinetic diameters of several commonly separated light gases. For instance, the separation of O₂ and N₂ from air is based on a size difference of 0.18 Å between the two molecules. Such a narrow size separation is challenging, especially from a materials development standpoint; even nanometer sized defects could compromise a membrane's performance. Defect-free membranes are essential to enable effective separation units.

Membrane performance is described in terms of specific gas permeability and selectivity and these metrics are commonly used to compare different membranes. The flux through dense polymers follows Fick's first law of diffusion where penetrant gases move through the membrane down a chemical potential gradient that

Table 1
Kinetic diameters and boiling points of industrially relevant gases.

Gas	He	H ₂	O ₂	N ₂	CO ₂	CH ₄
Kinetic Diameter (Å)	2.60	2.89	3.46	3.64	3.30	3.80
Boiling point (K)	4.23	20.37	90.25	77.36	216.6	111.66

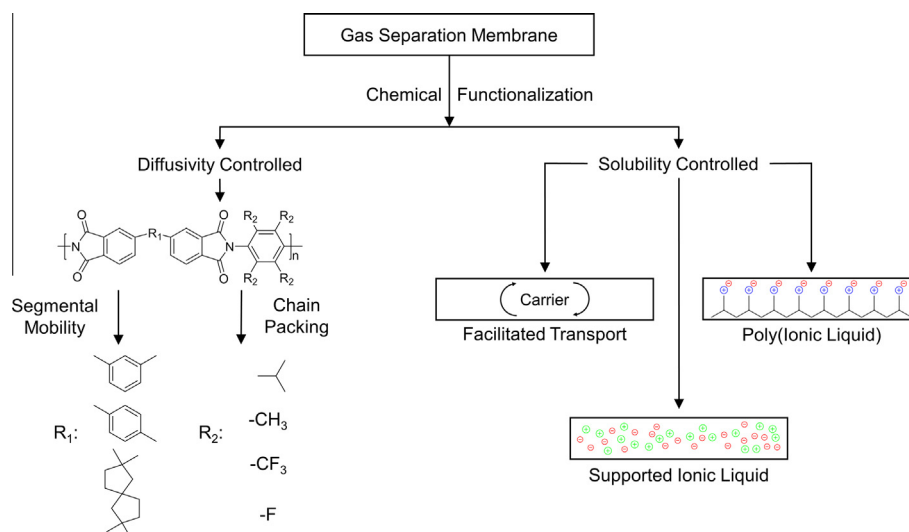


Fig. 1. The primary dichotomy of polymer functionalization discussed in this review.

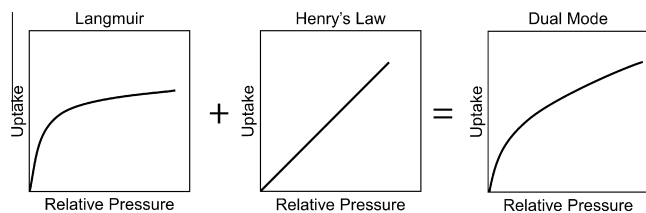


Fig. 2. The contribution of Langmuir type sorption at low pressures and Henry's Law type sorption at high pressures typical of dual mode sorption in glassy polymers.

can be expressed in terms of pressure, temperature, or concentration. The permeability, P , of a gas in a membrane is the flux, J , normalized by the membrane thickness, l , and the pressure differential across it, Δp , as given by Eq. (1). It is an intrinsic property of the material and its processing history that expresses the ease at which a gas molecule can move through a polymer.

$$P = \frac{J \cdot l}{\Delta p} \quad (1)$$

The permeability can be further broken down into its diffusivity, D , and solubility, S , contributions, as shown in Eq. (2). This equation is valid for a solution-diffusion model of gas transport through the membrane. In solution-diffusion, the gas must first sorb into the membrane on the upstream side, diffuse through the material, and then desorb on the downstream side.

$$P = D \cdot S \quad (2)$$

Typically only two of the parameters in the solution-diffusion model need to be measured experimentally, with the third calculated from Eq. (2). In common practice, the steady state permeability is measured and time-lag analysis is used to determine the diffusion coefficient from the transient region [4]. For very small gases with high diffusivities like H_2 and He, the time-lag is too small to accurately determine the diffusion coefficient. More accurate diffusivity coefficients can be determined by independently measuring the permeabilities and obtaining solubilities from sorption isotherms [5].

The diffusion coefficient is a measure of the speed (i.e., the mean squared displacement per unit time) a gas moves through the polymer due to random thermal motions. Diffusion typically follows an Arrhenius-type dependence on temperature and can vary by several orders of magnitude due to modifications to the polymer structure. The dramatic change in gas diffusivity by relatively simple chemical modifications has been the main focus of glassy polymer membrane separation research. Diffusion is fastest

for small molecules, hence diffusive selectivity favors smaller kinetic diameter molecules.

The solubility coefficient reflects the gas's solubility in the polymer and typically follows a van 't Hoff relationship with temperature. Changing the gas solubility is much more difficult than the diffusivity (especially for glassy polymers), and as such, chemical modifications only result in roughly an order of magnitude change in values. For light gases, solubility selectivity tends to favor the most condensable gas, i.e., the gas with the highest boiling point/critical temperature. Table 1 lists the boiling points of several commonly separated gases.

The most prevalent model of pressure dependent gas sorption in glassy polymers is the dual mode sorption model. At low pressures, the concentration of the gas in the polymer resembles a Langmuir type sorption response [6]. As the pressure increases, the rate of gas uptake begins to follow Henry's law—similar to a gas in a liquid [6]. The Langmuir isotherm is a result of the gas adsorbing in the excess free volume sites within the polymer. As shown in Eq. (3) and graphically in Fig. 2, dual mode sorption can be expressed as the algebraic sum of the Henry's law, C_D , and Langmuir, C_H , contributions. The Henry's law coefficient is commonly given by k_D , the Langmuir capacity constant by C_H' , and an affinity constant by b .

$$C = C_D + C_H = k_D p + \frac{C_H' b p}{1 + b p} \quad (3)$$

Different methods have been proposed to correlate gas permeability with polymer structure. Frequently the polymer free volume, and more specifically the fractional free volume (FFV), is related to the diffusion coefficient and permeability by Arrhenius-type relationships [7]. The concept of free volume is the difference in specific volume at equilibrium of a solid polymer and its occupied volume under normal conditions, a non-equilibrium quantity. Free volume can be easily visualized by comparing the packing of a low free volume polymer (crystalline polyethylene) and a high free volume polymer (PIM-1) as shown in Fig. 3. Clear voids are present between the PIM-1 chains that represent the instantaneous free volume. Over time, the polymer free volume will decrease due to the thermal motions of the chains relaxing towards their equilibrium packing.

Different theoretical models can be used to determine an occupied volume for a system. The most commonly used expression is 1.3 times the sum of the van der Waals volume, V_w , derived from Bondi's group contribution theory [8]. This term is the inverse average packing density at 0 K for complex organic molecules, which were used as polymer analogues [8]. Eq. (4) is used to estimate the FFV of a polymer where V_f is the free volume, V_{sp} is the specific volume measured from a sample, and V_{oc} is the occupied volume calculated based on theory.

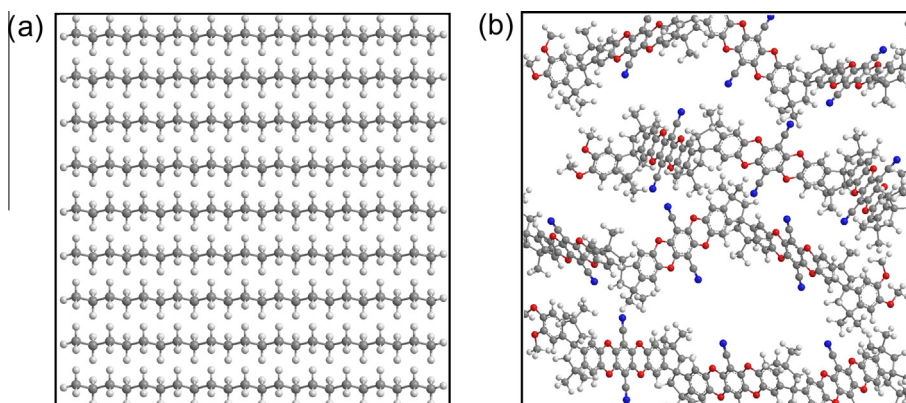


Fig. 3. Example of free volume in (a) crystalline polyethylene and (b) PIM-1.

$$\text{FFV} = \frac{V_f}{V_{\text{sp}}} = \frac{V_{\text{sp}} - V_{\text{oc}}}{V_{\text{sp}}} \quad (4)$$

In addition to permeability, selectivity is the other major defining property of a membrane. The ideal selectivity—also referred to as the permselectivity—of gas A compared to gas B, $\alpha_{A/B}$, is the ratio of single gas permeabilities, as given by Eq. (5). Similarly it can be expressed as a product of the diffusivity and solubility selectivities.

$$\alpha_{A/B} = \frac{P_A}{P_B} = \frac{D_A}{D_B} \frac{S_A}{S_B} \quad (5)$$

The majority of membrane research reports values in ideal selectivities. For non-interacting gases, this does not prove to be problematic and serves as a useful indication of the membrane's true performance. Permeation experiments with mixed gases typically try to simulate a feed stream of interest (e.g., 50% CO₂/50% CH₄, 14% CO₂/86% N₂, etc.). Mixed gas permeation data are much more representative of actual membrane performance since they account for the (primarily) competitive adsorption and (rarely) diffusional effects between the gases and polymer. In the case of mixed gas experiments, a separation factor, given by Eq. (6), is used rather than an ideal selectivity to describe the membrane's performance, although a selectivity based on the mixed gas permeabilities may still be calculated by Eq. (5). The y values correspond to the permeate composition and the x values to the feed.

$$\alpha_{A/B}^* = \frac{y_A/y_B}{x_A/x_B} \quad (6)$$

Although real process streams are composed of mixtures of many different gases, experimental setups are usually limited to binary systems when assessing membrane performance. Permeation studies of gas mixtures with more than two components quickly becomes mathematically unwieldy. Expressions for the diffusion coefficient become highly complex due to the number of paired interactions between the gases. However, a membrane's true performance ability is best understood when investigated under the specific conditions it will be utilized.

A strong interaction between gas penetrants and a polymer matrix that results in increased polymer segmental mobility is known as plasticization. Plasticization exhibits a general increase in permeability of all species and lowering of selectivity due to a disproportionate increase in the slow gas permeability. These changes are usually due to a species strongly sorbing to the polymer and causing it to swell, similar to a solvent.

Ideally, a membrane requires a high permeability combined with a high selectivity to make it attractive for industrial applications. This tradeoff is easily visualized using selectivity-permeability plots usually referred to as Robeson plots. These plots compare gas pair selectivity to single component permeability. A current state of the art line (typically referred to as Robeson's upper bound) is then drawn based on published ideal selectivities and permeabilities, with the upper right hand corner having the most desirable combination of properties. Newly synthesized membranes are often compared on Robeson plots as a quick gauge of feasibility. The initial plots created in 1991 were updated in 2008 to reflect the trends in improvement of gas separation membranes and have also been theoretically supported [9]. Only the 2008 upper bound lines will be used to compare different polymers in this review. These plots have, for the most part, remained the same since their inception except for three major cases. Polymers of intrinsic microporosity (PIM)s have shifted the CO₂/CH₄ upper bound, fluorocarbon polymers have dramatically shifted the He/CH₄, He/CO₂, and He/H₂ upper bounds, and thermally rearranged (TR) polymers exceed the new CO₂/CH₄ upper bound [10]. Thermally rearranged polymers were not considered when updating the upper bound line since they produce insoluble or only

partially soluble compounds, more akin to molecular sieving materials than true polymers [10].

3. Diffusivity controlled modifications

Chemical modifications to the polymer that seek to manipulate polymer chain packing are referred to as diffusivity controlled functionalization strategies. These polymer modifications look to introduce substitutions—either to the polymer chain backbone or side groups—to improve the gas separation performance of the material. These variations aim to change the diffusivity of the components by altering the polymer structure. Different design principles have been discussed to improve the selectivity without significantly lowering the permeability. The majority of gas separation membrane research falls under this category, which is attractive due to both the wide variety of options available to tailor the polymer for different applications and the immense variability in possible gas diffusion coefficients through polymer membranes.

3.1. Fundamental design principles

The prevailing idea for improving the performance of glassy polymers is to inhibit intersegmental chain packing while simultaneously hindering backbone mobility [11]. Rubbery polymers, on the other hand, behave somewhat differently than glassy polymers. They have very open structures that lend them high permeabilities without much size or shape selectivity, analogous to liquids [12]. The design principles discussed here are not usually used for rubbery polymers because of this fundamental difference in material behavior.

Structure–property relationships of polymers are an important tool to predict the separation performance of a membrane. The permeability in glassy polymers has been strongly correlated to its fractional free volume (FFV). Higher FFV materials typically have greater gas permeabilities than their lower FFV counterparts. Different methods can be employed to inhibit both segmental mobility and tight chain packing: adding bulky side groups to exploit steric interactions, using polar substituents, or changing the orientation of the linkages. We will give a brief overview of these basic design rules before moving on to state-of-the-art functionalization strategies.

Introducing large bulky side groups inhibits close chain packing, thereby increasing the FFV [12]. Expanding the amount of transient void space (i.e., free volume) in the polymer increases the frequency of diffusive jumps for the penetrant gas and thus increases its diffusivity. This favors the smaller gas relative to larger penetrants. However, the solubility selectivity decreases because both gases are soluble in the non-selective transient free space [13].

Using polar substituents like halogens to modify the polymer backbone has a similar effect as adding bulky side groups. The replacement of hydrogen with larger fluorine molecules increases the FFV, which also increases the permeability and solubility, but at the cost of selectivity [14]. When halogens are used to replace methyl groups instead of hydrogens, the permeability decreases while the selectivity increases [12]. In this case, the substitution of halogens increases the interchain attraction and motion without significantly altering the size of the substituent [12]. Introducing a halogen such as bromine next to a flexible linkage inhibits some chain rotation around that point and can help mitigate the selectivity loss associated with adding a bulky group; further functionalization can also be achieved at these reactive groups [15]. Other polar groups, not just halogens, can increase chain interactions. For instance, polyamide-imides (Torlon®) have been shown to have

impressive plasticization resistance due to both intra- and inter-chain hydrogen bonding between amide groups on the polymer backbone [16].

Switching the linkage on benzyl rings from para- to meta- (e.g., Fig. 1, R₁) results in the familiar tradeoff of decreased permeability and increased selectivity. The meta-linked polymers no longer have symmetric subunits and experience greater steric hindrance to chain rotation [12]. As shown in Appendix A.1, changing the benzyl ring linkage between a 6FDA polyimide and a hexafluoro diamine from para- to meta-results in a 47% increase in O₂/N₂ permselectivity, but an order of magnitude loss in permeability [17]. Asymmetric substitutions (such as placing a single ortho-methyl group on each benzyl ring of a polycarbonate) can also be used to modify the structure of a polymer, though these results vary between polymer families and substitution location. Symmetric substitutions usually favor an increase in both the permeability and selectivity by improving the chain stiffness while still maintaining a fairly open matrix. However, asymmetric substitutions are more likely to increase the selectivity at the cost of permeability by reducing the FFV [18]. The use of asymmetric substitutions, as well as meta/para substitutions, need to be used carefully to avoid dramatic losses in permeability [19].

Backbone stiffening reduces the amount of rotational freedom of the polymer chain. Highly flexible backbones allow for more compact packing of the polymer chains, which can negate the effects of adding large substituents. Stiffer backbones reduce the speed at which the polymer relaxes into its equilibrium state where there is essentially no FFV. Larger molecules are also better able to exploit the chain mobility than smaller molecules when making diffusive jumps, perhaps by forcing themselves between the polymer chains [12]. The simultaneous implementation of these two ideas has been suggested as an effective means of improving the membrane separation properties. *Using a single technique—introducing bulky groups without also increasing the chain rigidity or vice versa—does not produce the desired effects and diffusivity selectivity will decrease rather than increase* [12].

The basic design principles discussed above used to modify and improve polymer membranes has not changed dramatically since the early 1990s [12]. Current research has shifted towards the use of more exotic polymers, such as those containing perfluorinated groups and many various copolymers. Copolymerization opens up the possibilities for many different types of chemical blending and structure–property relationships. The resulting copolymers usually have properties between the two parent polymers and the monomer ratio can be used to fine tune the desired membrane. Polyimides in particular have been the subject of intense research for many different separations, ranging from light gases to hydrocarbons. Early research focused on varying the polyimide structure, while later work explored variations of hexafluoro-dianhydride (6FDA) based polyimides and different copolyimides [20]. The effects of different structural changes on the membrane performance of polyimides can be compared in Appendices A.1–A.3. An extensive table of permeation data for more traditional polymer families already exists and can be found elsewhere [12].

3.2. Ladder polymers

The most recent shift in the diffusivity control of polymers is the use of monomers containing severely motion-restricted segments. These restricted segments have no rotational degrees of freedom and frustrate chain packing. An example of a polycarbonate with a contortion site is shown in Fig. 4. The center carbon connecting the two cyclopentane rings, known as the spirocenter, is covalently bound to four different carbons. The inclusion of this monomer significantly increases the permeability relative to other

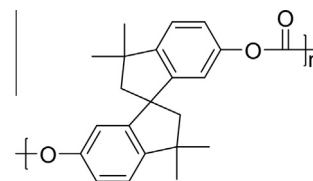


Fig. 4. Repeat unit of spirobisindane polycarbonate.

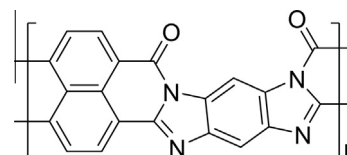


Fig. 5. Repeat unit of the ladder polymer poly(benzimidazobenzophenanthroline) BBL.

modified polycarbonates [21]. However the selectivity is only moderate for O₂/N₂ separations and quite poor for CO₂/CH₄ due to the large decrease in solubility selectivity [22].

Ladder polymers are another subset of polymers composed of rigid planar molecules with an uninterrupted sequence of rings [23]. These polymers have highly rigid backbones that make them potentially useful for gas separation membranes. An example ladder polymer poly(benzimidazobenzophenanthroline) (BBL) shown in Fig. 5 has a series of conjugated rings that gives rise to a highly rigid backbone; however there are no bulky, out-of-plane groups to disrupt the intersegmental chain packing. As a result, BBL packs efficiently into flat sheets and has a much lower FFV than some other semi-ladder polymers [24]. The resulting permeability and selectivity do not place BBL near the upper bound line for O₂/N₂ separations [24]. This further emphasizes the previously mentioned need for both bulky substituents and chain rigidity to produce effective membranes.

3.3. Polymers of intrinsic microporosity

Within the last decade, a new class of polymers with extraordinarily high surface area has been discovered by the Budd and McKeown research groups. Known as polymers of intrinsic microporosity (PIM)s, these polymers exhibit high permeabilities and selectivities for common gas pairs. An attractive feature of PIMs is that they are soluble in common solvents and membranes can be easily solution cast from them. By far the most well studied PIM is PIM-1, which is considered the prototypical PIM (shown in Fig. 6a). It is often compared to other highly permeable polymers like poly(trimethyl-silyl-propyne) (PTMSP). The membrane performance of PIM-1 for common gases is well documented by Budd et al. with detailed permeation data available in Appendices

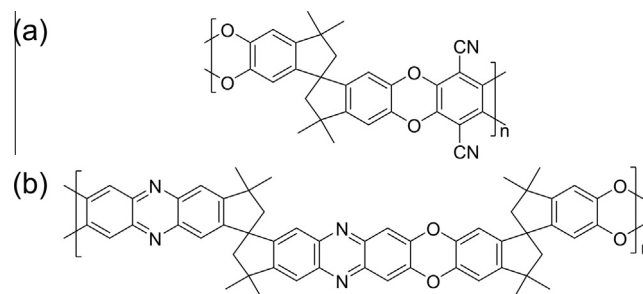


Fig. 6. Structures of (a) PIM-1 and (b) PIM-7.

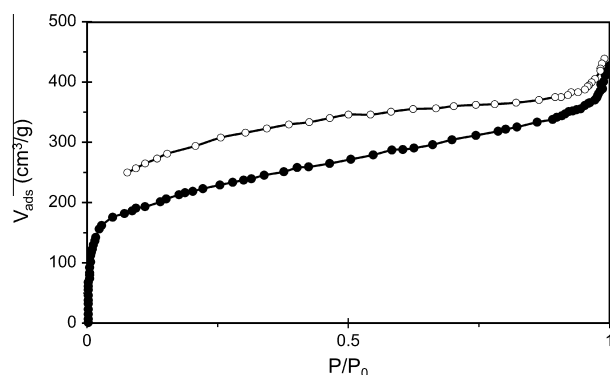


Fig. 7. PIM-1 nitrogen adsorption (filled circles) and desorption (open circles) isotherms at 77 K [29].

A.4–A.6 [25]. Even for hydrocarbon feeds like *n*-butane and methane mixtures, PIM-1 exhibits better performance than PTMSP due to its improved resistance to aging [26]. Despite this, PIMs exhibit some chain flexibility and age significantly compared to traditional gas separation membranes.

The synthesis of PIM-1 follows a single step polycondensation from commercially available monomers and reagents. More PIMs, such as PIM-7 (Fig. 6b) and PIM-8 through PIM-10, have been synthesized and characterized, but to a lesser extent than PIM-1; they all share the same design strategy of rigid units and frustrated chain packing via awkward, non-planar backbone units [27]. The different gas permeabilities of these PIMs are tabulated in detail in Appendices A.4–A.6. The versatility of monomers available for reaction allows for a wide variety of different PIMs to be synthesized and modified.

Both of the PIMs shown in Fig. 6 are characterized by the highly strained spirocenter and dioxane linkages that give these PIMs their quintessential behavior. PIMs have very high surface areas—850 m²/g for PIM-1—that are larger than most other polymers and even many zeolites [28]. The isotherm shown in Fig. 7 is close

to a Type I isotherm and reflects the microporosity of PIM-1 [29]. The high uptake at low pressures in the physisorption isotherm is indicative of micropores in the material; some mesoporosity is noticed as well [28]. The desorption hysteresis is apparently due to pore network effects where smaller size-sieving pores control access to larger pores [28].

The Pinnau and Guiver groups have actively pursued chemical functionalization of PIM backbones. Modifications to PIMs fall under two broad categories: (i) copolymerization with PIM monomers and (ii) post-polymerization modification at the cyano group. The addition of functionalized comonomers, usually bulkier groups, increases the selectivity while significantly decreasing the permeability. Du et al. have focused on copolymerizing different large monomers into the structure of PIM-1, including tetrafluoromethyl, phenylsulfone, disulfone, dinaphthyl, and thianthrene [30–33]. Bezzu et al. have also investigated the addition of a large spirobifluorene unit into PIM-1 [34]. These large groups increase the size (i.e., diffusion) selectivity of the membrane without significantly altering the solubility selectivity.

Post-polymerization modification at the cyano group allows for different functionalities to be introduced into the PIM structure without altering the polymerization mechanism. This method of modification looks to increase the membrane's affinity towards a specific gas. Fig. 8 illustrates a few examples of this PIM-1 functionalization and its effects on the selectivity/permeability trade-off. Commonly, the cyano group is changed to an amine or other nitrogen containing group to increase the CO₂ solubility and introduce hydrogen bonding to the system as an effort to improve the CO₂/CH₄ and CO₂/N₂ selectivity [35]. Du et al. and Swaidan et al. have added other functional groups like amidoximes, carboxylic acids, tetrazoles, and methyl tetrazoles to the cyano group [36–39]. Polar groups change the polymer's solubility in solvents as well; these modified PIMs are now only soluble in polar aprotic solvents [39].

While the high permeabilities and selectivities of PIMs make them attractive for gas separations, their properties are highly dependent on their processing history. Details in the film casting conditions can change the properties by an order of magnitude.

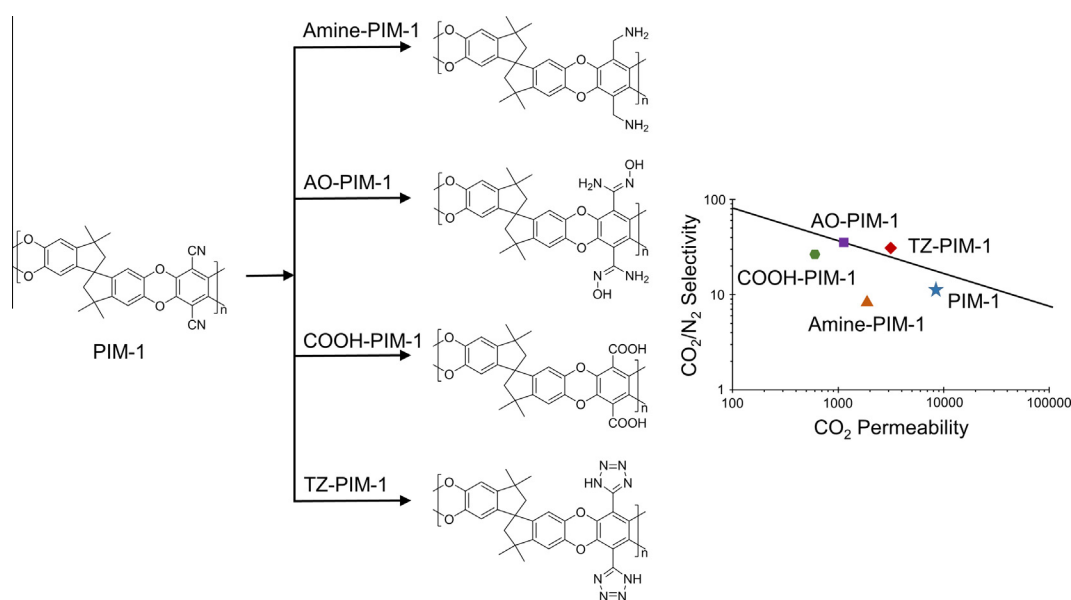


Fig. 8. Post-polymerization functionalization of PIM-1 at the cyano group and the effects on the CO₂/N₂ selectivity/permeability tradeoff. All films methanol-treated prior to permeation testing. PIM-1 measured at 25 °C and 50 psig feed pressure [37]. Amine-PIM-1 measured at 25 °C and 1 bar feed pressure [35]. Amidoxime-PIM-1 (AO-PIM-1) measured at 35 °C and 2 bar feed pressure [36]. Carboxylated-PIM-1 (COOH-PIM-1) hydrolyzed for 5 h at 120 °C and measured at 25 °C and 50 psig feed pressure [37]. Tetrazole-PIM-1 (TZ-PIM-1) at 70.5% conversion of the cyano group measured at 22 °C and 50 psi feed pressure [38]. Permeability given in Barrers; 1 Barrer = 10^{−10} cm³[STP] cm cm^{−2} s^{−1} cm Hg^{−1} = 3.35 × 10^{−16} mol m m^{−2} s^{−1} Pa^{−1}.

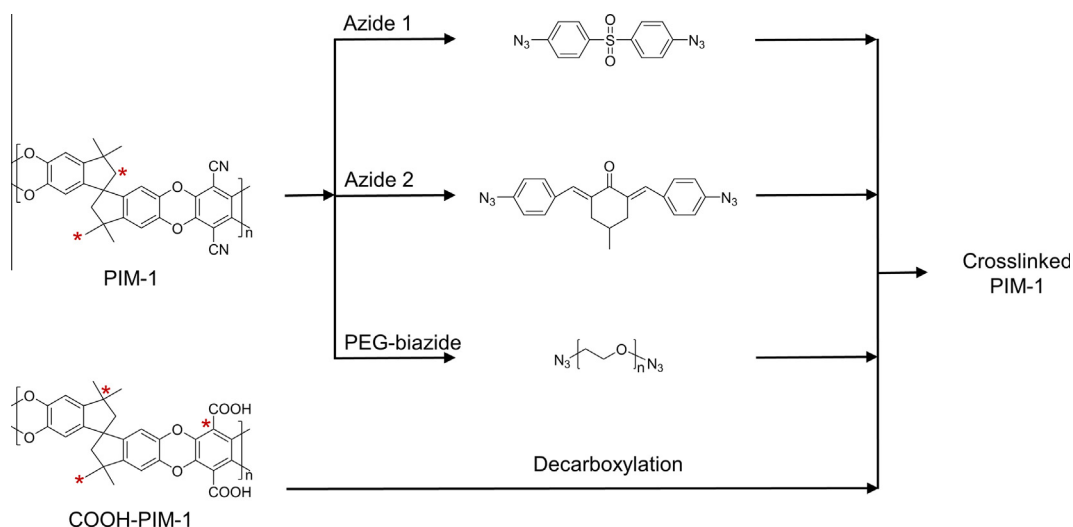


Fig. 9. Outline of nitrene-initiated diazide crosslinking and thermal decarboxylation crosslinking of PIM-1 [42–44]. Asterisks (*) indicate proposed crosslinking sites.

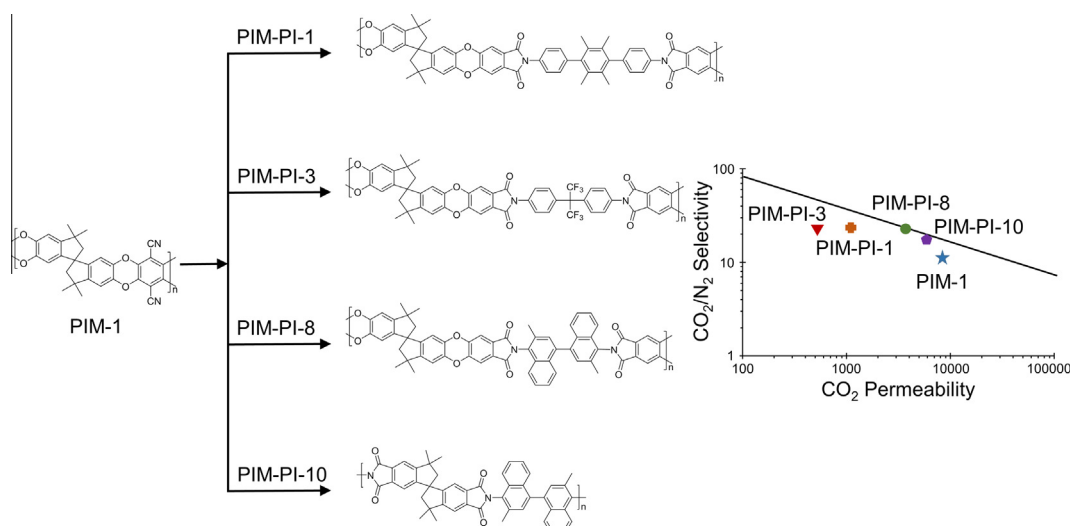


Fig. 10. Copolymerization of PIM-1 to form PIM-PIs and the effects on the CO₂/N₂ selectivity/permeability tradeoff. Methanol-treated PIM-1 film measured at 25 °C and 50 psig feed pressure [37]. PIM-PI-1, PIM-PI-3, and PIM-PI-8 films measured at 30 °C and 200–500 mbar feed pressure [45]. Methanol-treated PIM-PI-10 films measured at 30 °C and 1 bar [47]. Permeability given in Barrers; 1 Barrer = 10^{−10} cm³[STP] cm cm^{−2} s^{−1} cm Hg^{−1} = 3.35 × 10^{−16} mol m m^{−2} s^{−1} Pa^{−1}.

For PIM-1, contact with water decreases the permeability while soaking in methanol significantly increases the permeability [40]. This dependence on membrane history is likely responsible for the large variation in reported permeabilities and selectivities for PIMs and other highly permeable polymers [40]. These unusually strong dependencies can be used to improve membrane performance. For instance, methanol conditioning is used to essentially erase the thermal and processing history of PIM membranes as well as remove any residual casting solvent trapped in the pores [40]. Soaking highly permeable polymers in methanol swells the polymer and increases the chain mobility and free volume, thereby also increasing the permeability. Over time, the effects of the methanol treatment diminish—often interpreted as physical aging—as the polymer relaxes back towards its equilibrium state [41]. For a methanol-conditioned PIM-1 membrane left at ambient conditions, the rate of aging begins to stabilize after approximately five days; O₂ permeability decreases by 23% and N₂ permeability decreases by 40% after 45 days [40]. Even though PIMs are more resistant to fast aging

than other highly permeable polymers like PTMSP, they are not immune to the effects.

To overcome CO₂-induced plasticization, various crosslinked PIMs have been developed by Du et al. The highly reactive nitrene radical can be formed in situ and used to crosslink PIM-1 with different azides [42]. While crosslinking usually lessens the effects of plasticization at high CO₂ pressures, it significantly reduces the permeability to all gases with only a moderate increase in selectivity. In the case of C₃H₆/C₃H₈ separations, the introduction of a relatively long PEG-biazone crosslinker decreases the C₃H₆ permeability by 96% and increases the selectivity by 23% [43]. PIM-1 can also be functionalized with crosslinkable groups. Thermal decarboxylation of carboxylated PIM-1 is sufficient to increase the CO₂ plasticization resistance, but with the same selectivity/permeability tradeoff [44]. An outline of these different crosslinking schemes is given in Fig. 9.

Finally, PIMs have been modified by copolymerization with polyimides to form PIM-polyimide membranes. Several examples of PIM-PI copolymers that have been synthesized are shown in

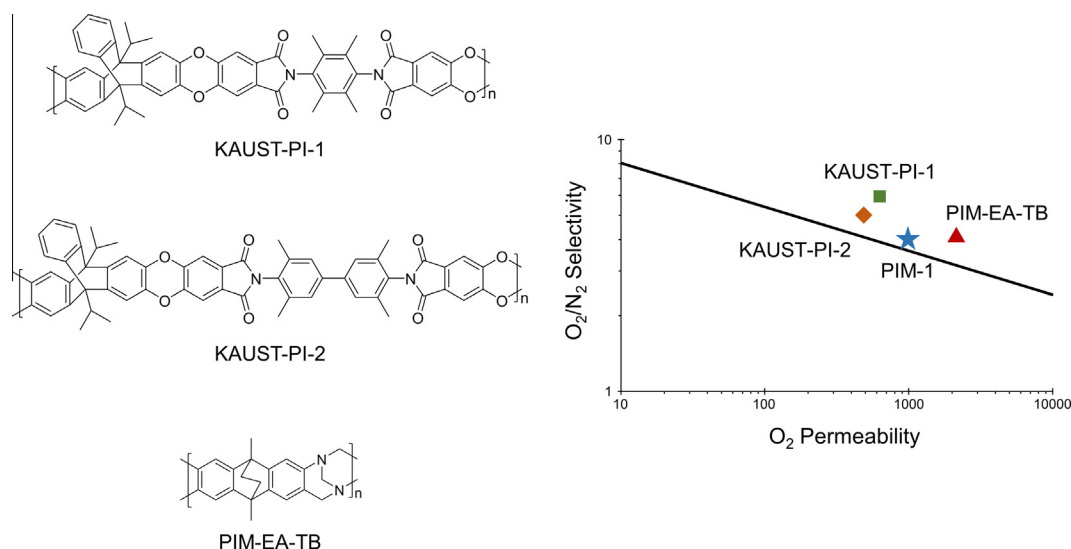


Fig. 11. Chemical structures of KAUST-PI-1, KAUST-PI-2, and PIM-EA-TB and their O_2/N_2 performance. All films methanol-treated prior to permeation testing. PIM-1 measured at 35 °C and 2 bar feed pressure [36]. PIM-EA-TB measured at 25 °C and 1 bar feed pressure [49]. KAUST-PI-1 and KAUST-PI-2 measured at 35 °C and 2 bar feed pressure [48]. Permeability given in Barrers; 1 Barrer = $10^{-10} \text{ cm}^3(\text{STP}) \text{ cm cm}^{-2} \text{ s}^{-1} \text{ cm Hg}^{-1} = 3.35 \times 10^{-16} \text{ mol m m}^{-2} \text{ s}^{-1} \text{ Pa}^{-1}$.

Fig. 10; more examples are listed in detail in Appendices A.4–A.6. The characteristic hindered packing of the PIM section gives PIM-PIs higher surface areas and permeabilities compared to pure polyimides [45]. The gas selectivity of these polymers also follows the preferences of the PIM rather than the polyimide; the somewhat mobile PI segment allows for slightly less frustrated chain packing, thus driving up the PIM-PI selectivity relative to neat PIMs. Similarly, the use of PIM-PIs improves the gas separation properties to levels higher than that of pure polyimides [46].

The spirocenter of PIM-1 and its derivatives is not completely rigid due to the presence of the single carbon joining the cyclopentane rings that can exhibit some flexibility. In order to further reduce the chain mobility and improve selectivity, Ghanem et al. have synthesized a series of triptycene based PIM-PIs (KAUST-PIs) [48]. The site of contortion in these polymers, shown in Fig. 11, is a non-planar, bridged tricyclic molecule that provides additional rigidity to the structure while inhibiting close chain packing [48]. Ortho-methyl groups are used to prevent rotation around the imide bonds, further hindering chain mobility [48]. These modifications dramatically improve the membrane performance to levels that far exceed the upper bound for O_2/N_2 , H_2/N_2 , and H_2/CH_4 [48]. Moreover, these polymers demonstrate that a fine balance can be struck between segmental mobility and chain packing; structures that are too open may exceed the upper bound, but have selectivities that are too low to be practical membranes.

The ease of modification and inclusion of other classes of polymers opens up many opportunities for PIMs to introduce microporosity to other polymer families. The polymers mentioned here further illustrate the utility of the previously mentioned design principles in improving a membrane's gas separation performance.

4. Solubility controlled modifications

Solubility controlled modifications (Fig. 1) look to introduce functionality that manipulates the solubility coefficients of specific gases in the polymer to influence permeability. These types of modifications are inherently different than diffusivity controlled modifications since they exploit a variety of chemistries and are not limited to substitutions along the polymer chain. Solubility controlled modifications are usually aimed at acid gases (such as

CO_2 and H_2S) and other gases that can chemically react or interact (such as olefins) with different side groups since size sieving is not emphasized. Although the scope of research for solubility controlled modifications is much narrower than diffusivity controlled modifications, the inherent difference between mechanisms can be utilized to better understand polymer functionalization holistically.

4.1. Facilitated transport

Facilitated transport membranes are a class of membranes that combine traditional polymeric membranes with carrier agents to improve selectivity and are one of the more developed solubility controlled membranes for gas separation. The main method of transport is the reaction between the reactive gas and carrier to form a complex that then moves through the membrane. All gases are still subject to the usual solution-diffusion mechanism. A schematic of facilitated transport through a membrane is given in Fig. 12. The carrier-gas complexation occurs reversibly; ideally the process can cycle and operate continuously without the need for additional carrier.

Several defining characteristics set facilitated transport membranes apart from the glassy amorphous polymeric membranes that dominate the field. These membranes can be highly selective, have high desired gas permeabilities relative to other membranes

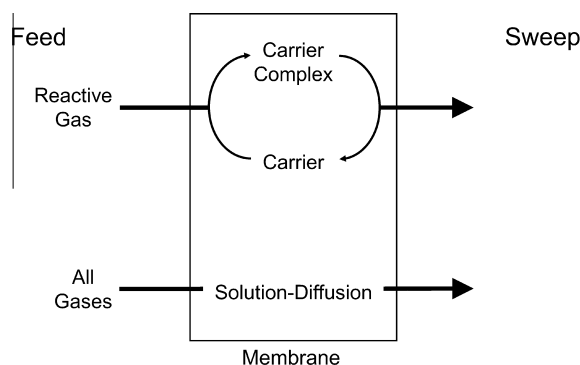


Fig. 12. Diagram of gas transport through a facilitated transport membrane.

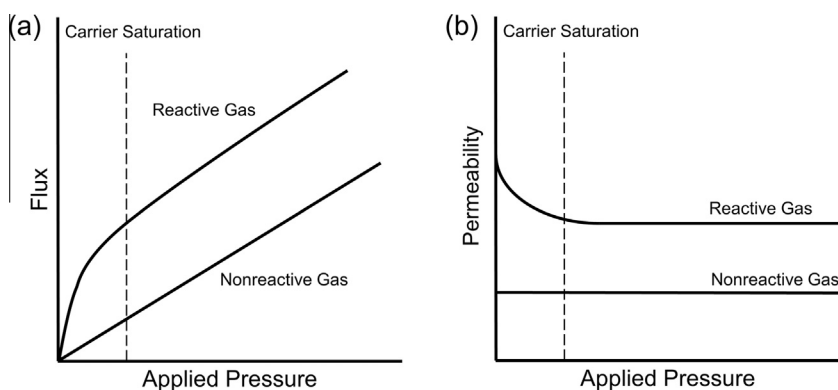


Fig. 13. The influence of facilitated transport on (a) the flux and (b) the permeability of different types of gases.

at low driving forces, and only require small amounts of complexing agent; however, they can be somewhat unstable [50]. Due to the presence of the carrier, transport is controlled by both the complexation reaction as well as the chemical potential gradient across the membrane.

Unlike traditional glassy polymers, facilitated transport membranes operate best under conditions with low driving forces. A low upstream driving force reduces the permeability of all gases; this allows for the complexing agent to be the most effective pathway for the desired penetrant gas relative to the solution-diffusion pathway. The carrier complex only increases the permeation of the reactive gas and at low pressures all of the carriers are ideally being utilized. At these low pressures, a high selectivity can be achieved. Indeed, as the partial pressure of the desired penetrant increases, the permeability and selectivity decrease concomitantly. This phenomenon is known as carrier saturation [50]. At high partial pressures, the complexing agent becomes saturated with gas and its contribution becomes negligible relative to solution-diffusion. Under these conditions, the selectivity drops off significantly because of the high driving force of the non-reactive/undesirable gas. The influence of both transport mechanisms gives these membranes unique flux and permeability curves as shown in Fig. 13. These properties can be utilized for separations that are traditionally very difficult for other polymer systems—namely dilute separations.

Modern facilitated transport membranes originate from liquid membranes. Although liquid membranes for gas separations will not be discussed in detail in this review, it is necessary to mention their importance in relation to polymeric chemically modified membranes. Traditional liquid facilitated transport membranes are composed of a carrier-filled layer of fluid. Unfortunately, liquid membranes are less stable than their solid polymeric counterparts; the solvent can evaporate, accumulate impurities, or even blow out of the support during operation—the physical separation of the liquid from the support due to high pressure differentials.

Facilitated transport membranes can be separated into three main types, listed in order of increasing stability: immobilized liquid membranes, solvent-swollen polymer membranes, and solid polymer membranes with reactive functional groups. All of these membranes look to improve upon the stability issues of liquid membranes while maintaining the associated high selectivity and permeability properties.

The immobilized liquid membrane—also called a supported liquid membrane (SLM)—is the least stable of the facilitated transport membranes and was the first attempt to overcome the limitations of emulsion liquid membranes by simply changing the operating configuration. Immobilized liquid membranes have the liquid carrier carefully held in a microporous support by capillary forces where the separation occurs. These membranes are

prone to drying out even under optimal conditions. Polymeric hollow fiber supports improve upon some of these stability issues. This configuration reduces the solvent loss during operation due to a tighter pore structure and therefore increases the mechanical stability of the system [51]. Recently, research has focused on using ionic liquids (ILs) as the membrane material in SLMs due to their unique physical and chemical properties, although ILs have been around for over a century [52]. Ionic liquids have negligible vapor pressure, high thermal stability, high ionic conductivity, and are nonflammable [53]. Despite these improvements, supported ILs are still subject to accumulating impurities that are even more difficult to remove due to the nature of the fluid. Even at moderate pressure differentials—less than 2 atm—SLMs are subject to liquid blow out that limits the overall feasibility of the configuration.

Solvent-swollen membranes, or mobile carrier membranes, are intermediate in stability between supported liquid membranes and solid polymeric membranes. The solvent causes the membrane to swell, allowing the liquid carrier solution to be distributed throughout the entire system [51]. The gas then absorbs into the liquid where it can react with the carrier to form a complex and proceed with the normal facilitated transport mechanism. Since these types of membranes still rely on a liquid carrier, they are prone to some of the same limitation as immobilized liquid membranes. Commonly, ion-exchange membranes have been used to retain the carrier by electrostatic interactions and reduce carrier loss into process streams [54].

The most stable facilitated transport membrane is the fixed-site carrier membrane. Fixed-site carrier membranes are similar to glassy polymer membranes as a result of the carrier being covalently attached to the polymer. Since the carrier is not required to be a liquid solution, these membranes cannot dry out and overcome many of the limitations that plague other facilitated transport membranes. While the exact mechanism of gas transport within the system has not been fully elucidated, it is commonly explained as the gas molecule jumping between fixed-sites along the polymer. In the case of a combined fixed-site and mobile carrier membrane, the mobile carrier assists the reactive gas in moving between fixed-sites. Several different theories have been discussed that involve the reactive gas moving between fixed carriers in the polymer matrix. Different models have been used to analyze facilitated transport, both in the diffusion limited regime and the reaction limited regime [55]. The analytical solutions to these models can be used to quickly identify the feasibility of a potential carrier for use with a membrane.

The applications of facilitated transport membranes fall into distinct categories. The first major focus is for oxygen facilitated transport membranes. Free metal coordinated complexes within the membrane, such as Co(II) coordinated to porphyrins or Schiff bases, have been investigated [56,57]. These carriers, shown in

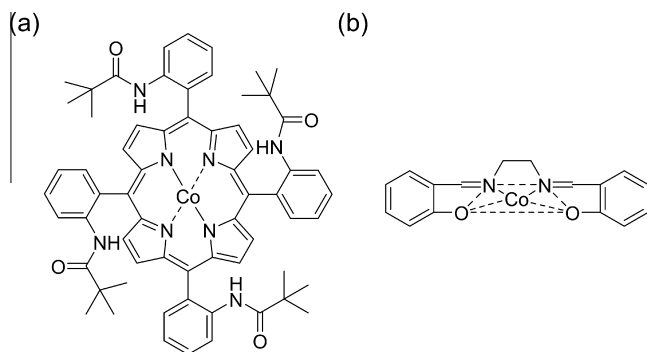
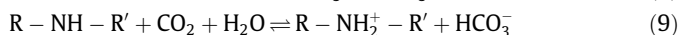
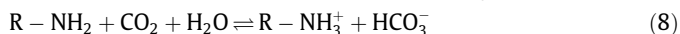


Fig. 14. Metal coordinated carriers, (a) cobalt porphyrin and (b) cobalt Schiff base, used to transport oxygen.

Fig. 14, are coordinated to the polymer matrix and mimic oxygen transport of hemoglobin in blood [58]. Metal carriers hinder large scale production of oxygen and oxygen-enriched air from facilitated transport membranes due to concerns over metal contamination in the product streams and carrier oxidation. Stability issues associated with the carrier and film integrity have prevented any major implementation of oxygen facilitated transport membranes. The next major focus is olefin/paraffin separations, in part due to the large potential for energy savings compared to cryogenic distillation. Similar to carriers used for oxygen transport, metal ions like silver and copper can bind reversibly to the olefin to create a carrier complex, as shown in Fig. 14. Silver ions (Ag^+) outperform both Cu^+ and Cu^{2+} in olefin permeability and olefin/paraffin selectivity [59]. Large weight percentages, up to 67 wt%, of carrier can be added to these membranes without a loss in mechanical stability, although the chemical stability is still problematic [60]. Not only can the Ag^+ degrade in the presence of impurities, but it can form highly reactive byproducts like silver acetylide salts.

The last major separation area investigated using facilitated transport membranes is acid gas removal used to produce commercial product streams and meet emission standards. Carbon dioxide and other acid gases (e.g., H_2S) readily react with basic functional groups. The Ho group has worked on using poly(allylamine) and 2-aminoisobutyric acid as the fixed-site and free carriers respectively, due to the presence of free amine groups in these materials [61–63]. Recently, sterically hindered amine polymers have been shown to increase CO_2 transport more so than unhindered amines [64]. The reactions between unhindered amines (Eqs. (7) and (8)), hindered amines (Eq. (9)), and potassium carbonate (Eq. (10)) with humid carbon dioxide shows some different carrier possibilities in a membrane [65].



In all of these amine-containing membranes, the presence of water is highly influential to CO_2 transport because it allows for multiple reaction pathways to occur and can act as an additional mobile carrier to further improve the membrane's performance [66]. At higher temperatures, the membrane presumably retains less water and exhibits a decrease in both CO_2 selectivity and permeability [61,64]. These amine-containing membranes, more so than other facilitated transport membranes, are dependent on the quality and compositional fluctuations of the feed.

Unfortunately, the carrier is both the most beneficial aspect and the limiting factor towards successful utilization of facilitated transport membranes. An ideal carrier binds just strong enough to the reactive gas to maximize facilitation while not binding so

strongly as to hinder release of the gas downstream. In many cases, as the feed pressure increases, carrier saturation becomes a serious concern, especially when working with amine and carbonate carriers, further limiting the acceptable operating range of the membrane. Poisoning by feed impurities, unwanted side reactions, or simple wash-out from the membrane must all be considered when looking at the long term stability of the membrane.

Facilitated transport membranes offer an alternative method to remove acid gases from dilute feed streams. They are particularly effective at low driving forces and have also demonstrated some use for oxygen enrichment and the selective transport of olefins. These membranes perform best at room temperatures and pressures to avoid drying out the membrane and prevent carrier saturation. Their limitations make them somewhat difficult to scale up and operate, but they have an advantage in dilute or low pressure applications where other membranes fail to perform well.

4.2. Poly(ionic liquid)s

Another emerging group of solubility controlled modified membranes is poly(ionic liquid)s. Ionic liquids are defined as salts with weak coordination bonds that give them melting points below 100°C . An important subsection of this class of materials is the ILs that remain as liquids at ambient temperatures. These room temperature ionic liquids, or RTILs, remain liquid over a wide temperature range, some as large as -96°C to 300°C [67]. Poly(ionic liquid)s, or PILs, are solid polymers derived from ILs that share many of their physical and chemical properties. The use of PILs as gas separation membranes originated from the need for more stable liquid membranes not susceptible to blow out. A SLM composed of a solid PIL support with an IL membrane has very strong ionic interactions with the polymer support, creating a solvent-swollen membrane that does not rely solely on capillary forces. These membranes have been shown to withstand pressure differences that surpass other SLMs without any sign of solvent loss [68].

Poly(ionic liquid)s are synthesized by attaching IL components to easily polymerizable monomers—frequently styrene, because it can be readily polymerized through different methods. Polymerization occurs almost exclusively through the base polymer attached to the cation (which dictates the separation selectivity) leaving the anion free within the system. The structures of a few common PIL cations and anions are shown in Fig. 15. In general, large resonance-stabilized anions are preferred over smaller anions because they do not bind as strongly to the cation [69]. Tang et al. have synthesized PILs with various cations. The PIL relative CO_2 sorption capacities follow the same trends as their IL counterparts. Membranes made out of ammonium-based ionic monomers have much higher CO_2 sorption capacities than imidazolium PILs due

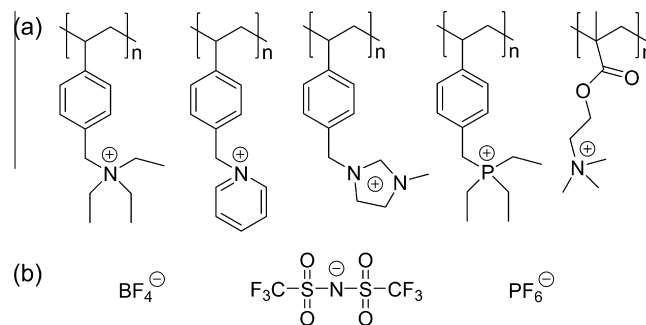


Fig. 15. (a) Common cations used as PILs listed in order from left to right. Poly[1-(para-vinylbenzyl)-triethylammonium]; poly[1-(para-vinylbenzyl)pyridinium]; poly[1-(para-vinylbenzyl)-4-methylimidazolium]; poly[1-(para-vinylbenzyl)triethylphosphonium]; poly[2-(methacryloyloxy)ethyl]. (b) Common anions used with PILs listed in order from left to right. $[\text{BF}_4]^-$ = tetrafluoroborate; $[\text{TF}_2\text{N}]^-$ = trifluoromethanesulfonamide; $[\text{PF}_6]^-$ = hexafluorophosphate.

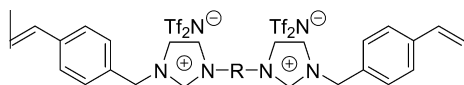


Fig. 16. General structure of a cross-linkable GRTLIL monomer [75].

to the presence of the amine group [70]. Even though the CO_2 solubilities are relatively low, imidazole-based ILs and PILs are by far the most well studied due to the simplicity of the base ion, ease of modification, and wide variety of chemistries available.

Similar to facilitated transport membranes, PIL membranes separate molecules based on differences in the gas's solubility in the polymer rather than its diffusivity through it. For a variety of ILs and PILs, the CO_2 sorption is significantly higher than either CH_4 or N_2 due to Lewis acid-base interactions between the CO_2 and nitrogen-containing groups [71]. Moreover, the CO_2 absorption in RTILs has been shown to be fast and completely reversible [70]. These membranes combine the desired properties of ILs with the improved stability, durability, and processability associated with polymers [72]. Thus far the bulk of PIL membrane research has focused on the separation of CO_2 from CH_4 for natural gas sweetening and CO_2 from N_2 for purifying post-combustion flue gas.

Pure PIL membranes have predominantly been investigated for CO_2 selective separations. Bara et al. have focused on developing relationships between cation/anion functionality and membrane performance; Since PILs used for gas separations consist of a neutral backbone with the IL on the side chain, the choice of polymer backbone only slightly alters the permeability and selectivity [73]. Instead, the permeability and selectivity properties of PILs are controlled by the choice of IL group. The inclusion of polar groups on the cation have been shown to increase the CO_2/CH_4 and CO_2/N_2 separation factors by about 50% over their alkyl chain analogues [74]. These polar groups within the polymer interact more strongly with the CO_2 and enhance the selectivity. In the extreme case of very long alkyl chains, the ionic character of the PIL becomes negligible compared to the alkyl chain interactions and membrane selectivity is lost. Adding functionality to the IL component can be used to fine tune the selectivity and permeability of a membrane, which can be observed in detail in Appendices A.7–A.9. In general, PIL membranes for CO_2/N_2 separation are seen as more favorable than for CO_2/CH_4 separation because of the higher selectivities that can be achieved.

Bara et al. have also investigated the effects of crosslinking PILs. The introduction of gemini crosslinked imidazoles, shown in Fig. 16, dramatically reduces the gas permeability of CH_4 and N_2 to the level of a barrier material rather than a membrane [75]. The highly crosslinked system creates a very tortuous diffusion path for the gas. The large gemini structure of the IL monomer

significantly alters the physical properties relative to the single RTIL. Only the use of the very large Tf_2N^- anion (Fig. 15b, middle) allows for the pre-polymerized IL salt to remain a liquid at room temperature [75]. The degree of crosslinking, rather than the specific functional groups, dominates the permeation properties that ultimately restricts gas diffusion and reduces the permeability [75]. In this system, the CO_2/CH_4 separation is more favorable than the CO_2/N_2 separation, contrary to behavior found in almost all other PIL membranes to date [75].

Placement of the IL group in relation to the polymer chain can also be modified. Fig. 17 compares the ionic charge placement in several PIL configurations. All the PILs mentioned so far have consisted of hydrocarbon backbones with pendant IL side groups. A main chain polyimidazolium ionene was synthesized and compared to a similar side chain imidazole PIL; the polyimidazolium ionene synthesized by Carlisle et al. exhibited a CO_2 permeability on par with other PILs [76]. Although the separation performance did not exceed those of previously synthesized PILs (Appendices A.7–A.9), it does expand the possibilities of functionalizing PILs. Fig. 18 compares the CO_2/N_2 separation performance of several representative solubility controlled membranes.

Polymerizing the ILs dramatically reduces the permeability of the system while moderately increasing the selectivity. In order to increase the polymer permeability back to levels comparable to ILs, composite PIL-IL membranes have been created. The CO_2 permeability of an IL is approximately two orders of magnitude greater than a PIL. Typically a mixture of 80%/20% polymerizable/nonpolymerizable ILs are photopolymerized to create a solvent-swollen composite membrane [77]. The liquid acts as a plasticizer to increase the permeability of the membrane while maintaining CO_2 selectivity as the polymer adds more diffusional selectivity than a nonselective porous support. Applied pressures up to 2 atm do not exhibit any signs of liquid blow out, unlike previously developed SLMs [68]. Incorporation of as little as 20% IL into a membrane has been shown to increase the permeability of an imidazolium-based PIL by almost 400% over the corresponding neat polymer [77].

In these PIL-IL composite membranes, the substituents of the IL incorporated into the membrane are known as the free cation and free anion respectively. Although the presence of the free cation significantly increases the permeability of a PIL, the specific functional groups on them do not. Having a free cation in the membrane has a much greater effect on the CO_2 permeability and selectivity—relative to a neat polymer—than any individual change to the functional groups on the free cation itself [78]. Since the increase in permeability due to the IL is not entirely dependent on the functionalization, the IL acts more as a plasticizer to the polymer membrane [78]. The separation remains dominated by the polymer matrix rather than the free liquid. Differences between anions have a weaker influence on the CO_2 absorption than functionalization of the cation. The functional groups on the cation can interact with the CO_2 (which affects the diffusivity and solubility) while it appears that the anion size, rather than functionality, predominantly influences the separation. Bara et al. found that changing the anion from the commonly used Tf_2N^- introduces a more difficult anion exchange and purification during synthesis for little gain in selectivity [68]. The large Tf_2N^- free anion affords the highest permeability with similar selectivities to other anions tested [68].

Another method to impart the improved physical durability of a solid to an IL is the use of a gelling agent. Voss et al. found that the addition of 1.5 wt% low molecular weight organic crosslinker can form a gel with IL solutions [79]. This gel improved the mechanical stability and had a higher burst pressure (the pressure required to force the gel to separate from the support) compared to the analogous IL while maintaining a similar gas separation performance [79].

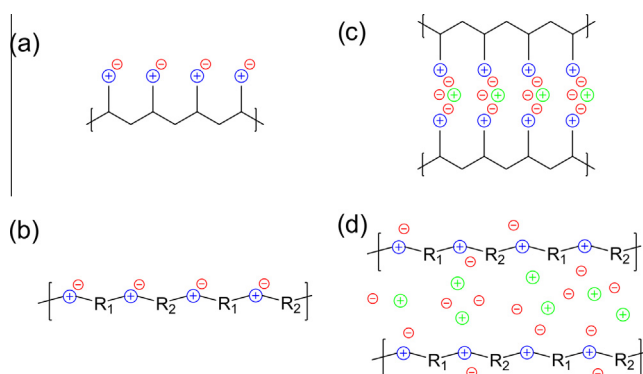


Fig. 17. Generic structures of a (a) PIL, (b) ionene, (c) PIL-IL composite, and (d) ionene-IL composite. Reproduced with permission from [76].

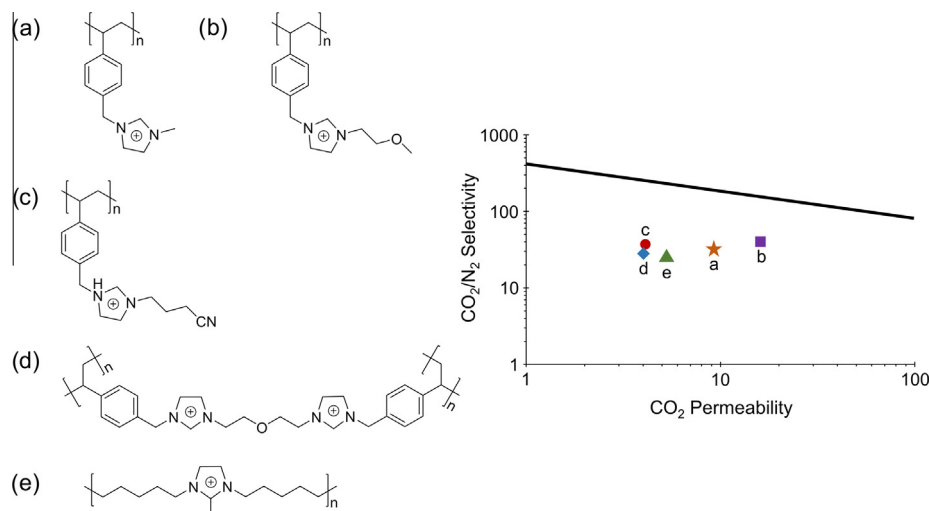


Fig. 18. Imidazolium-based PILs and the effect on the CO_2/N_2 selectivity/permeability tradeoff due to (a) methyl [73] (b) oligo(polyethylene glycol) [74] (c) alkyl-terminated nitrile [74] (d) PEG crosslinked gemini [75] (e) ionene [76]. All of the neat polymers shown have $[\text{TF}_2\text{N}]$ anions. Permeability given in Barrers; 1 Barrer = $10^{-10} \text{ cm}^3[\text{STP}] \text{ cm cm}^{-2} \text{ s}^{-1} \text{ cm Hg}^{-1} = 3.35 \times 10^{-16} \text{ mol m m}^{-2} \text{ s}^{-1} \text{ Pa}^{-1}$.

The benefits of using PIL membranes for gas separations are that they combine desirable CO_2 absorption and physical properties of ILs with the structural stability of solid polymers. These membranes overcome many of the limitations that hinder facilitated transport membranes. Unlike glassy polymers, PILs exhibit reversible CO_2 -induced plasticization [80]. Presumably these polymers are flexible enough to return to their original state after desorption of the plasticizing agent, while glassy polymers are too rigid and remain highly permeable but nonselective. These membranes can also be restored to their initial state by simply running a pure feed stream through the membrane, eliminating the need for more complicated regeneration steps [80].

Several limitations to PILs should be noted. While ILs have very high thermal stabilities, PILs are not necessarily as robust. It is possible for PILs to have polymer backbones that depolymerize before the pendant groups decompose or react [81]. Ionic liquids are also good solvents for a variety of compounds. Especially in the case of supported liquid and composite membranes, the accumulation of impurities should be considered when discussing long term stability. Indeed, ILs are not easily purified by distillation because of their negligible vapor pressure. Moreover, the highly hydrophilic nature of ILs makes control over water accumulation in the membrane difficult. As of now, it is uncertain how the presence of water will affect the permeability and selectivity of these membranes. Finally the stability in harsh feeds should be investigated before any scale up can be considered.

4.3. Surface treatments

Using exotic functional groups to chemically modify polymeric membranes improves the separation performance, but at a price. Modifying the entire polymeric membrane with functional groups can be problematic to the material's processability, depending on the type of functional groups used. Functionalized polymers can become insoluble in common solvents, extremely costly to produce, or have other undesirable material properties. Another method used to impart functionality without sacrificing processability is surface treatments of the polymers, shown in Fig. 19. Functionalization of the membrane can take the form of polymerizing already functionalized monomers, crosslinking the base polymer, or adding functional groups to the surface via plasma treatments [82–85]. Especially for relatively expensive functional groups—such as fluorine—surface functionalization can be an

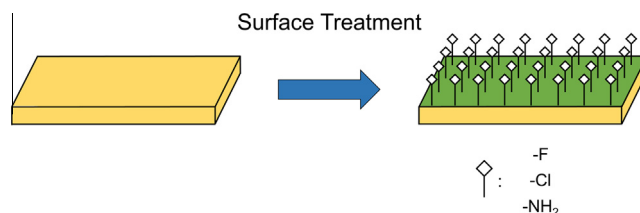


Fig. 19. Membrane functionalization via plasma polymerization to modify the surface chemistry. Surface treatments can add a thin layer of functionalized polymer (with pendant functional groups) to an existing membrane.

attractive method to improve the membrane's selectivity [86]. Functionalization of just the surface can be less costly to fabricate while still significantly improving the membrane's performance. A wide variety of different functionalization strategies and methods have been used to alter polymer membranes for different separations [87]. Especially in asymmetric membranes where only a thin layer performs the separation, surface modifications could be beneficial to improving the selectivity without dramatically increasing the processing difficulty or cost associated with membrane fabrication.

4.4. Reverse-selective membranes

All of the polymers discussed so far exhibit behaviors that follow the Robeson upper bound—decreasing selectivity with increasing single component permeability. However under certain circumstances, the opposite behavior can also be observed. These membranes, known as reverse-selective membranes, go against the traditional notion of the small molecule diffusing fastest through the membrane; the large molecule actually has the highest permeability. These reverse-selective membranes have apparent upper bound lines that increase in both selectivity and permeability of the large molecule. Mixed gas selectivity is usually higher than single component permselectivity in these systems as well.

Despite the general notion that all plasticization is detrimental to membrane performance, it can sometimes be utilized to improve the selectivity. To achieve reverse-selectivity, the membrane needs to have a high solubility selectivity while maintaining a low diffusivity selectivity—an area where plasticization can potentially be beneficial [88]. Plasticization increases the diffusivity of the large molecule to a greater extent than the small molecule, thereby reducing the

importance of the diffusivity selectivity and allowing the solubility selectivity to dominate permeation. Commonly rubbery polymers are used because they contain flexible oxygen linkages, such as ethers, in the repeat unit that can increase the solubility of more reactive gases like CO₂.

Reverse-selective membranes have been investigated for hydrogen production, purification of flue gas, and natural gas sweetening. Pinnau et al. systematically modified different poly(acetylenes) and investigated their gas separation performance for upgrading hydrocarbon feeds. Reverse-selective behavior was observed for several isopropyl-based poly(dialkylacetylenes) when separating propane from nitrogen and n-butane from methane [89]. Especially in polymeric membranes, hydrocarbons are more soluble than permanent light gases in the polymer and cause the membrane to swell. This swelling—just as with CO₂-induced plasticization—reduces the diffusivity selectivity, allows the solubility selectivity to dictate the final permeability selectivity, and results in reverse-selective behavior. Membranes made from reverse-selective materials also deviate in other aspects typically associated with glassy polymers. For different side chain length poly(2-alkylacetylenes), the permeability does not correlate with the FFV. As the length of the alkyl chain increases, the FFV decreases due to the more flexible chains packing efficiently, yet the permeability of most gases increases—showing a greater importance in chain mobility than FFV [90].

Reverse-selective membranes are an interesting deviation from most polymers used for gas separations. The utility of reverse-selective membranes is generally limited to separations that involve removing a more condensable gas from a relatively inert gas—such as organic vapors from nitrogen or methane. They differentiate themselves from the design principles used to tailor glassy polymers. Typical membranes have the small molecule as the permeate, which can require costly recompression before use. Reverse-selective membranes can potentially avoid these costs by keeping the smaller molecule as the high-pressure retentate.

4.5. Rubbery polymers

Although not the main focus of this review, rubbery polymers are used as solubility selective membranes for applications often involving acid gases like CO₂, and thus warrant a brief discussion. Rubbery polymers behave somewhat differently than glassy polymers and as such the previously mentioned design principles are not always effective. Similar to reverse-selective membranes, rubbery polymers separate gases by exploiting differences in gas solubility within the polymer. A common rubbery polymer used for CO₂ separations is poly(ethylene oxide), or PEO. This polymer contains flexible ether linkages that increase the solubility of CO₂ relative to permanent light gases. Depending on the molecular weight and temperature, PEO can contain varying levels of crystallinity. Crystalline segments within polymers are practically impermeable to gases and are generally avoided as a result. Different strategies have been used to control the amount of crystallinity in PEO membranes, such as copolymerization with less flexible polymers and blending [91]. These PEO-containing membranes are usually hydrophilic and present new challenges in forming ultra-thin or asymmetric membranes that could improve their industrial appeal. The amount of variability in mixing these polymers also allows for a wide range of different polymers to be formed. While these membranes perform best at low temperatures and pressures, polymer crystallization becomes an important factor to consider. Operating conditions must be optimized to balance high gas solubility—and therefore selectivity—with realistic feed conditions for a given process. According to Liu et al., for the separation of CO₂ from flue gas, plasticization might be beneficial to expanding the useable range of the membrane because of the reverse-selective nature of these polymers [91].

5. Summary and perspectives

Engineering membranes for gas separations through functional polymer chemistry is a vibrant and productive field of research. Here we have identified two general strategies used by the community to create designer membranes for a variety of industrially relevant gas separations: (i) backbone functionalization with the intent of controlling the relative rate of diffusive jumps by gas pairs within a membrane and (ii) functionalization with the intent of controlling a targeted gas's reactivity or solubility within the membrane.

The use of diffusivity controlled modifications is diverse and usually only limited by the chemistry involved in the synthesis. The most beneficial improvements to the selectivity are achieved by simultaneous inhibition of close chain packing and backbone segmental mobility. The variety of options available to modify a polymer's diffusivity make it an attractive method to improve the membrane performance and is currently the simplest and most common method employed. Research has shifted away from functionalizing relatively simply polycarbonates and polysulfones to modifying more exotic materials such as fluorinated polyimides, a variety of different copolymers, and PIMs. The reactions needed to modify polymer diffusivity are fairly easy to scale up and can show significant effects on the permeability and selectivity with only minor changes to the original structure. A major concern that must be considered is the solubility and mechanical properties of the resulting modified polymer. Especially with crosslinked polymer membranes, the polymer may become insoluble in common organic solvents or extremely brittle, properties that severely limit the processability and therefore utility of the membrane. Significant effort has been directed towards creating ultra-rigid polymer backbones (e.g., PIM-1), which pack inefficiently and result in high FFV. Despite being near or above the Robeson upper bound, these membranes are most likely too open and not selective enough for demanding gas separation applications. As a result, there has been a very recent refocusing on the balance of chain packing and segmental mobility to drive up the selectivities of these membranes. Indeed, it seems likely that emerging diffusion controlled engineered polymers will combine hindered moieties such as spirocenters, triptycenes, or barrelenes with segments such as functionalized polyimides to promote chain packing.

While not as well developed as diffusivity controlled membranes, solubility controlled modifications look to improve gas separations by exploiting different chemical interactions between the membrane and the gas. Facilitated transport membranes work best at low driving forces, an area that most other membranes fail to efficiently separate gases. However, even trace quantities of contaminants (H₂S, SO₂, etc.) can permanently poison the active carriers; there has been no breakthrough solution of this issue to date. Ionic liquids and poly(ionic liquid)s overcome many of the stability issues that have plagued the earlier facilitated transport membranes. In spite of this, the accumulation of impurities and other forms of carrier deactivation are still detrimental to long term stability. Continued research to improve this stability while maintaining the high gas permeabilities and selectivities of these membranes and development of high surface area membrane configurations will be the most beneficial to future large scale implementation. Specific surface modifications also offer another method to tailor a membrane towards a desired application. The use of expensive specialty chemicals as the carrier can be an issue at an industrial scale and should be taken into consideration. Although the feasibility is not well investigated, solubility controlled membranes have the potential to be useful in situations that diffusivity controlled polymers are not.

Covalent organic frameworks (COFs) are the polymer equivalents to metal organic frameworks (MOFs). These polymers have

long range order and permanent porosities on the same level as MOFs while exhibiting low densities and high thermal stabilities [92]. Research on COFs is in its infancy with most of the work focused on synthesis, although initial applications look to be for gas storage, catalysis, and optoelectronic devices. The pore size and packing structure can be modified based on the material choice. Although the application of COFs for gas separation membranes has not been investigated yet, it is a new research area that has the potential to lead to polymer-based membrane materials with well-defined pore sizes capable of molecular sieving separations similar to MOFs, ZIFs, and zeolites.

Finally, there seems to be significant potential to combine both functionalization strategies to find a middle ground capable of taking advantage of the strengths of both. This hybrid approach will initially find the most utility in gas separations that have opposing selectivity mechanisms; CO₂/H₂S/CH₄ (sour gas sweetening) is a prime example. To remove both acid gases from methane requires a membrane capable of diffusive selectivity for CO₂ over CH₄ and solubility selectivity for H₂S over CH₄. Kraftschik et al. have very recently

demonstrated a combination of diffusion controlled and solubility controlled functionalization by crosslinking an advanced polyimide (6FDA-DAM:DABA (3:2)) with poly(ethylene glycol) chains [93]. The resulting membrane is able to achieve industrially relevant selectivities for both acid gases over methane. We expect that new materials following this approach will be developed. A major challenge facing this hybrid approach is balancing the high chain mobility of the solubility enhanced segments with the chain packing enhancing and segmental mobility reducing segments. Continued research towards the development of design rules and structure–property relationships for gas separation membranes will ultimately lead to new, energy efficient membrane units capable of augmenting and ultimately replacing existing inefficient separation processes.

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Appendix A

A.1. Permeability and permselectivity for selected polyimides¹

Polymer abbreviation	Permeability (Barrer)						Permselectivity					Temperature (°C)	Pressure (atm)	Reference
	He	H ₂	O ₂	N ₂	CO ₂	CH ₄	He/N ₂	H ₂ /N ₂	O ₂ /N ₂	CO ₂ /N ₂	CH ₄ /N ₂			
6FDA-APPS	–	–	1.90	–	9.10	0.30	–	–	–	–	–	35	1	[94]
6FDA-6FmDA	–	–	1.80	0.26	–	–	–	–	6.9	–	–	35	2	[17]
6FDA-6FpDA	–	–	16.30	3.47	–	–	–	–	4.7	–	–	35	2	[17]
6FDA-DAM	533.21	–	191.53	54.99	842.41	46.81	9.7	–	3.5	15.3	0.9	35	6.803	[95]
6FDA-mPDA	80.46	–	5.39	0.87	20.25	0.35	92.5	–	6.2	23.3	0.4	36	6.803	[95]
6FDA-DABA	53.24	–	3.14	0.51	12.80	0.20	104.4	–	6.2	25.1	0.4	37	6.803	[95]
6FDA-DAM: DABA (3:2)	195.91	–	31.99	7.04	143.95	4.19	27.8	–	4.5	20.4	0.6	38	6.803	[95]
6FDA-DAM: mPDA (3:2)	238.93	–	52.08	12.46	230.21	7.94	19.2	–	4.2	18.5	0.6	39	6.803	[95]
6FDA-mPDA: DABA (3:2)	42.74	–	2.51	0.40	9.14	0.16	106.9	–	6.3	22.9	0.4	40	6.803	[95]
6FDA-DAT	–	81.96	8.83	1.53	39.59	0.69	–	53.6	5.8	25.9	0.5	35	1.974	[96]
6FDA-DATRI	198	257	39	8.10	189	6.20	24.4	31.7	4.8	23.3	0.8	35	1	[97]
6FDA-durene	355.40	585.40	134.80	38.70	677.80	33.70	9.2	15.1	3.5	17.5	0.9	35	2	[98]
6FDA-m-DDS	–	–	0.73	0.08	2.30	0.03	–	–	9.1	28.7	0.4	35	3	[99]
6FDA-p-DDS	–	–	3.70	0.61	16	0.34	–	–	6.1	26.4	0.6	35	3	[99]
6FDA-PI-CH	–	23.10	1.92	0.33	9.64	–	–	70	5.8	29.2	–	22.5	1.261	[100]
6FDA-PI-M	–	21.30	2.05	0.32	5.58	–	–	66.6	6.4	17.4	–	22.5	1.261	[100]
6FDA-TAPA- AH-ODA	–	27	–	0.25	6.70	0.11	–	108	–	26.8	0.4	35	1	[101]
6FDA-TAPA- AM-0.15EG	–	32	–	0.50	16	–	–	64	–	32	–	35	1	[101]
6FDA-TAPA- AM-0.25TPA	–	86	11	2.16	65	1.59	–	39.8	5.1	30.1	0.7	35	1	[101]
6FDA-TAPA- AM-0.34EG	–	22	1.90	0.33	11	0.22	–	66.7	5.8	33.3	0.7	35	1	[101]

¹ Detailed permeation data for traditional polymer families can be found in [12].

Appendix A (continued)

Polymer abbreviation	Permeability (Barrer)						Permselectivity					Temperature (°C)	Pressure (atm)	Reference
	He	H ₂	O ₂	N ₂	CO ₂	CH ₄	He/ N ₂	H ₂ / N ₂	O ₂ / N ₂	CO ₂ / N ₂	CH ₄ / N ₂			
6FDA-TAPA-AM-1.4EG	–	–	–	0.12	3.70	0.08	–	–	–	30.8	0.7	35	1	[101]
6FDA-TMPDA	–	499.90	110.35	30.13	555.72	24.49	–	16.6	3.7	18.4	0.8	35	1.974	[96]
6FDA-TMPDA: DAT (1:1)	–	182.56	26.81	5.71	130.21	3.35	–	32.0	4.7	22.8	0.6	35	1.974	[96]
6FDA-TMPDA: DAT (1:3)	–	125.41	15.40	2.99	74.69	1.56	–	41.9	5.2	25	0.5	35	1.974	[96]
6FDA-TMPDA: DAT (3:1)	–	224.82	36.50	8.28	187.63	5.54	–	27.2	4.4	22.7	0.7	35	1.974	[96]
6F-HPI	67.85	42.75	3.73	0.55	16.95	0.21	123.4	77.7	6.8	30.8	0.4	25	1	[102]
BADBSBF-BBBPAn	–	–	52.20	12	–	–	–	–	4.4	–	–	25	2.904	[103]
BADBSBF-BTDA	–	–	18	2	–	–	–	–	9	–	–	25	2.904	[103]
BADBSBF-BTSBPAn	–	–	121	54	–	–	–	–	2.2	–	–	25	2.904	[103]
BADBSBF-FDA	–	–	28	12	–	–	–	–	2.3	–	–	25	2.904	[103]
B-HPI	23.57	14.26	0.72	0.09	2.74	0.03	261.9	158.4	8.0	30.4	0.3	25	1	[102]
BPDA-6FpDA	–	–	6.65	1.21	–	–	–	–	5.5	–	–	35	2	[17]
Br-Matrimid	29	–	3	0.46	14	0.42	63	–	6.5	30.4	0.9	35	1	[15]
Matrimid	22	–	1.90	0.27	8.70	0.24	81.5	–	7	32.2	0.9	35	1	[15]
BTDA-6FmDA	–	–	0.39	0.05	–	–	–	–	8.3	–	–	35	2	[17]
BTDA-6FpDA	–	–	1.86	0.31	–	–	–	–	6	–	–	35	2	[17]
Bt-HPI	19.75	11.13	0.60	0.08	1.43	0.02	246.9	139.1	7.5	17.9	0.3	25	1	[102]
C-PI-CH	–	9.20	0.91	0.22	1.68	–	–	41.8	4.1	7.6	–	22.5	1.261	[100]
C-PI-M	–	8.30	0.47	0.08	1.96	–	–	103.8	5.9	24.5	–	22.5	1.261	[100]
DSDA-TAPA-AH-DDS	–	6.1	–	0.02	1	–	–	254.2	–	41.7	–	35	1	[101]
DSDA-TAPA-AM-TPA	–	12	–	0.09	4.00	0.11	–	130.4	–	43.5	1.2	35	1	[101]
DS-PI-CH	–	10.20	0.75	0.16	3.20	–	–	63.8	4.7	20	–	22.5	1.261	[100]
DS-PI-M	–	12.70	0.91	0.16	3.60	–	–	79.4	5.7	22.5	–	22.5	1.261	[100]
E-PI-CH	–	13.23	0.90	0.08	3.76	–	–	165.4	11.3	47	–	22.5	1.261	[100]
E-PI-M	–	10.50	0.68	0.09	3.05	–	–	116.7	7.6	33.9	–	22.5	1.261	[100]
HFA/TERT	66.43	–	5.88	1.16	26.82	0.86	57.3	–	5.1	23.1	0.7	35	2	[104]
HFA/TERT-b-HFA/ISO 9	90.68	–	8.27	1.62	40.09	1.01	56	–	5.1	24.7	0.6	35	2	[104]
HFA/TERT-b-HFA/ISO 12	86.93	–	8.59	1.61	37.94	1.09	54	–	5.3	23.6	0.7	35	2	[104]
HFA/TERT-b-HFA/ISO 18	154.82	–	15.43	3.06	70.93	2.13	50.6	–	5	23.2	0.7	35	2	[104]
HFAISO	32.46	–	1.68	0.28	7.60	0.18	115.9	–	6	27.1	0.6	35	2	[104]
I-6F	–	–	9.34	7.85	229.23	8.92	–	–	1.2	29.2	1.1	35	2.902	[105]
I-DS	–	–	2.59	2.36	80.46	3.77	–	–	1.1	34.1	1.6	35	2.902	[105]
II-6F	–	–	5.76	4.87	149.72	6.11	–	–	1.2	30.7	1.3	35	2.902	[105]
II-DS	–	–	1.33	1.01	34.11	1.18	–	–	1.3	33.8	1.2	35	2.902	[105]
O-HPI	23.45	13.36	0.54	0.06	1.81	0.02	390.8	222.7	9	30.2	0.3	25	1	[102]
PBO-PI (0:10)	23.60	14.30	0.72	0.09	2.70	0.03	262.2	158.9	8	30	0.3	30	1.013	[106]
PBO-PI (2:8)	29.80	38.20	2.20	0.40	11.40	0.30	74.5	95.5	6	28.5	0.8	30	1.013	[106]
PBO-PI (5:5)	42.10	47.10	4.80	0.82	25.30	0.65	51.3	57.4	6	30.9	0.8	30	1.013	[106]
PBO-PI (8:2)	479	623	90	18	389	14	26.6	34.6	5	21.6	0.8	30	1.013	[106]
PBO-PI (10:0)	857	1228	220	48	1014	41	17.9	25.6	5	21.1	0.9	30	1.013	[106]
P-HPI	62.21	35.21	2.58	0.36	9.87	0.08	172.8	97.8	7	27.4	0.2	25	1	[102]
PI-CH	–	13.30	1.11	0.28	5.68	–	–	47.5	4	20.3	–	22.5	1.261	[100]
PI-M	–	8.80	0.57	0.11	2.51	–	–	80	5	22.8	–	22.5	1.261	[100]

A.2. Diffusivity and diffusivity selectivity for selected polyimides

Polymer abbreviation	Diffusivity (10^{-8} cm ² /s)					Diffusivity selectivity			Temperature (°C)	Pressure (atm)	Reference
	H ₂	O ₂	N ₂	CO ₂	CH ₄	O ₂ /N ₂	CO ₂ /N ₂	CH ₄ /N ₂			
6FDA-6FmDA	–	2.24	0.57	–	–	3.9	–	–	35	2	[17]
6FDA-6FpDA	–	12.50	3.98	–	–	3.1	–	–	35	2	[17]
6FDA-DAT	–	5.39	1.26	1.77	0.18	4.3	1.4	0.1	35	1.974	[96]
6FDA-DATRI	442	23	5.50	13	1.50	4.2	2.4	0.3	35	1	[97]
6FDA-durene	–	–	–	20.77	5.18	–	–	–	35	2	[98]
6FDA-m-DDS	–	1.10	0.16	0.41	0.02	6.7	2.5	0.1	35	3	[99]
6FDA-p-DDS	–	5.50	1.12	2.10	0.18	4.9	1.9	0.2	35	3	[99]
6FDA-TAPA-AM-0.15EG	–	–	0.74	0.93	–	–	1.3	–	35	1	[101]
6FDA-TAPA-AM-0.25TPA	–	9.40	2.06	2.70	0.31	4.6	1.3	0.2	35	1	[101]
6FDA-TAPA-AM-0.34EG	–	2.42	0.46	0.59	0.08	5.3	1.3	0.2	35	1	[101]
6FDA-TMPDA	–	36.31	13.65	16	3.57	2.7	1.2	0.3	35	1.974	[96]
6FDA-TMPDA: DAT (1:1)	–	13.53	3.51	4.73	0.67	3.9	1.3	0.2	35	1.974	[96]
6FDA-TMPDA: DAT (1:3)	–	8.35	2.11	3	0.35	4	1.4	0.2	35	1.974	[96]
6FDA-TMPDA: DAT (3:1)	–	17.29	4.84	6.50	1.04	3.6	1.3	0.2	35	1.974	[96]
6F-HPI	331	–	–	0.43	0.09	–	–	–	30	1	[102]
B-HPI	236	–	–	0.13	0.02	–	–	–	30	1	[102]
BPDA-6FpDA	–	6.98	1.68	–	–	4.2	–	–	35	2	[17]
BTDA-6FmDA	–	0.76	0.12	–	–	6.5	–	–	35	2	[17]
BTDA-6FpDA	–	2.19	0.55	–	–	4.0	–	–	35	2	[17]
Bt-HPI	144	–	–	0.08	0.02	–	–	–	30	1	[102]
DSDA-TAPA-AM-TPA	–	–	0.21	0.20	–	–	1	–	35	1	[101]
HFA/TERT	–	8.10	1.61	3.35	0.39	5	2.1	0.2	35	2	[104]
HFA/TERT-b-HFA/ISO 9	–	46.15	13.44	16.90	2.51	3.4	1.3	0.2	35	2	[104]
HFA/TERT-b-HFA/ISO 12	–	65.97	16.04	31.95	3.88	4.1	2	0.2	35	2	[104]
HFA/TERT-b-HFA/ISO 18	–	268.20	30.14	52.71	8.29	8.9	1.7	0.3	35	2	[104]
HFAISO	–	3.20	0.77	1.43	0.13	4.2	1.9	0.2	35	2	[104]
O-HPI	814	–	–	0.13	0.02	–	–	–	30	1	[102]
P-HPI	257	–	–	0.39	0.04	–	–	–	30	1	[102]

A.3. Solubility and solubility selectivity for selected polyimides

Polymer abbreviation	Solubility (cm ³ (STP)/cm ³ atm)					Solubility selectivity			Temperature (°C)	Pressure (atm)	Reference
	H ₂	O ₂	N ₂	CO ₂	CH ₄	O ₂ /N ₂	CO ₂ /N ₂	CH ₄ /N ₂			
6FDA-6FmDA	–	0.61	0.35	–	–	1.7	–	–	35	2	[17]
6FDA-6FpDA	–	0.99	0.67	–	–	1.5	–	–	35	2	[17]
6FDA-DAT	–	1.22	0.91	16.95	2.89	1.3	18.6	3.2	35	1.974	[96]
6FDA-DATRI	0.46	1.29	1.14	11.02	3.12	1.1	9.7	2.7	35	1	[97]
6FDA-durene	–	–	–	24.80	4.92	–	–	–	35	2	[98]
6FDA-m-DDS	–	0.51	0.39	4.26	1.12	1.3	–	–	35	3	[99]
6FDA-p-DDS	–	0.52	0.43	5.57	1.47	1.2	–	–	35	3	[99]
6FDA-TAPA-AM-0.15EG	–	–	0.52	13.68	–	–	26.5	–	35	1	[101]
6FDA-TAPA-AM-0.25TPA	–	0.87	0.80	18.24	3.88	1.1	22.9	4.9	35	1	[101]
6FDA-TAPA-AM-0.34EG	–	0.59	0.54	13.68	2.17	1.1	25.3	4	35	1	[101]
6FDA-TMPDA	–	2.28	1.67	26.37	5.24	1.4	15.8	3.1	35	1.974	[96]

Appendix A (continued)

Polymer abbreviation	Solubility (cm ³ (STP)/cm ³ atm)					Solubility selectivity			Temperature (°C)	Pressure (atm)	Reference
	H ₂	O ₂	N ₂	CO ₂	CH ₄	O ₂ /N ₂	CO ₂ /N ₂	CH ₄ /N ₂			
6FDA-TMPDA: DAT (1:1)	–	1.52	1.22	20.90	3.80	1.3	17.2	3.1	35	1.974	[96]
6FDA-TMPDA: DAT (1:3)	–	1.37	1.06	18.92	3.42	1.3	17.8	3.2	35	1.974	[96]
6FDA-TMPDA: DAT (3:1)	–	1.60	1.29	21.96	4.10	1.2	17	3.2	35	1.974	[96]
6F-HPI	0.0980	–	–	30.09	1.75	–	–	–	30	1	[102]
B-HPI	0.0456	–	–	16.57	0.99	–	–	–	30	1	[102]
BPDA-6FpDA	–	0.72	0.54	–	–	1.3	–	–	35	2	[17]
BTDA-6FmDA	–	0.39	0.31	–	–	1.3	–	–	35	2	[17]
BTDA-6FpDA	–	0.65	0.41	–	–	1.6	–	–	35	2	[17]
Bt-HPI	0.0585	–	–	13.68	1.06	–	–	–	30	1	[102]
DSDA-TAPA- AM-TPA	–	–	0.33	15.20	–	–	45.5	–	35	1	[101]
HFA/TERT	–	0.55	0.41	6.08	1.68	1.4	15.0	4.1	35	2	[104]
HFA/TERT- b-HFA/ISO 9	–	0.14	0.09	1.80	0.31	1.5	19.7	3.3	35	2	[104]
HFA/TERT- b-HFA/ISO 12	–	0.10	0.08	0.90	0.21	1.3	11.8	2.8	35	2	[104]
HFA/TERT- b-HFA/ISO 18	–	0.04	0.08	1.02	0.20	0.6	13.3	2.5	35	2	[104]
HFAISO	–	0.40	0.28	4.04	1.05	1.4	14.6	3.8	35	2	[104]
O-HPI	0.0122	–	–	10.41	0.76	–	–	–	30	1	[102]
P-HPI	0.1064	–	–	19.38	1.52	–	–	–	30	1	[102]

A.4. Permeability and permselectivity for selected ladder and semi-ladder polymers

Polymer abbreviation	Permeability (Barrer)						Permselectivity					Temperature (°C)	Pressure (atm)	Reference
	He	H ₂	O ₂	N ₂	CO ₂	CH ₄	He/ N ₂	H ₂ / N ₂	O ₂ / N ₂	CO ₂ / N ₂	CH ₄ / N ₂			
SBIPC	49	57	7	1.4	30	2	35	40.7	5	21.4	1.4	35	2	[21]
BBL	–	–	0.02	–	–	–	–	–	–	–	–	35	3	[24]
PBIBI-BPA	–	–	16.14	2.89	83.26	2.16	–	–	5.6	28.8	0.7	25	1	[108]
PBIBI-DOD	–	–	15.16	2.61	70.81	1.74	–	–	5.8	27.1	0.7	25	1	[108]
PBIBI-PPD	–	–	26.22	4.94	137.20	3.78	–	–	5.3	27.8	0.8	25	1	[108]
PBIBI-TDP	–	–	19.55	3.13	92.10	2.43	–	–	6.2	29.4	0.8	25	1	[108]
PIM-1 ^a	660	1300	370	92	2300	125	7.2	14.1	4	25	1.4	30	0.197	[25]
PIM-7	440	860	190	42	1100	62	10.5	20.5	4.5	26.2	1.5	30	0.197	[25]
Amine-PIM-1 ^b	1200	3070	895	230	1890	303	5.2	13.3	3.9	8.2	1.3	25	1.014	[35]
AO-PIM-1 ^b	412	912	147	33	1153	34	12.5	27.6	4.5	34.9	1	35	2.027	[36]
CARDO-PIM-1	170	320	59	13	430	22	13.1	24.6	4.5	33.1	1.7	30	0.198	[27]
DC-PIM1 ^b	–	–	554	161	2345	192	–	–	3.4	14.6	1.2	25	3.4	[44]
DC-PIM2 ^b	–	–	411	118	1987	157	–	–	3.5	16.8	1.3	25	3.4	[44]
DC-PIM3 ^b	–	–	375	103	1996	116	–	–	3.6	19.4	1.1	25	3.4	[44]
DC-PIM4 ^b	–	–	238	57.60	1536	86.40	–	–	4.1	26.7	1.5	25	3.4	[44]
DC-PIM5 ^b	–	–	231	49.90	1291	52.60	–	–	4.6	25.9	1.1	25	3.4	[44]
DNPIM-25	–	3049	1139	321	5799	–	–	9.5	3.5	18.1	–	25	4.401	[33]
DNPIM-33	–	2347	907	242	4646	–	–	9.7	3.7	19.2	–	25	4.401	[33]
DNPIM-50	–	1478	522	132	2627	–	–	11.2	4.0	19.9	–	25	4.401	[33]
DNTOTPIM-25	–	2695	973	287	4814	–	–	9.4	3.4	16.8	–	25	4.401	[33]
DNTOTPIM-33	–	2077	773	228	3693	–	–	9.1	3.4	16.2	–	25	4.401	[33]
DNTOTPIM-50	–	1703	621	172	3065	–	–	9.9	3.6	17.8	–	25	4.401	[33]
DSPIM1-33	–	–	322	88	1408	–	–	–	3.7	16.0	–	25	4.401	[31]
DSPIM2-33	–	–	216	52	1077	–	–	–	4.2	20.7	–	25	4.401	[31]
DSPIM3-33	–	–	369	93	2154	–	–	–	4.0	23.2	–	25	4.401	[33]
MTZ100-PIM	–	–	270	63	1391	–	–	–	4.3	22.1	–	25	4.401	[39]

(continued on next page)

Appendix A (continued)

Polymer abbreviation	Permeability (Barrer)						Permselectivity					Temperature (°C)	Pressure (atm)	Reference
	He	H ₂	O ₂	N ₂	CO ₂	CH ₄	He/N ₂	H ₂ /N ₂	O ₂ /N ₂	CO ₂ /N ₂	CH ₄ /N ₂			
TZPIM-1	–	–	–	89	2562	–	–	–	–	28.8	–	25	4.401	[38]
TZPIM-2	–	–	–	103	3108	–	–	–	–	30.2	–	25	4.401	[38]
TFMPSPIM1	–	–	156	33	731	–	–	–	4.7	22.2	–	25	4.401	[30]
TFMPSPIM2	–	–	308	75	1476	–	–	–	4.1	19.7	–	25	4.401	[30]
TFMPSPIM3	–	–	561	158	2841	–	–	–	3.6	18.0	–	25	4.401	[30]
TFMPSPIM4	–	–	737	217	3616	–	–	–	3.4	16.7	–	25	4.401	[30]
TOTPIM-25	–	3567	1596	570	6441	–	–	6.3	2.8	11.3	–	25	4.401	[33]
TOTPIM-33	–	2854	1247	436	5320	–	–	6.5	2.9	12.2	–	25	4.401	[33]
TOTPIM-50	–	2616	1198	413	4756	–	–	6.3	2.9	11.5	–	25	4.401	[33]
TOTPIM-100	–	1368	642	190	3056	–	–	7.2	3.4	16.1	–	25	4.401	[33]
KAUST-PI-1 ^b	1771	3983	627	107	2389	105	16.6	37.2	3.1	22.3	1	35	2.027	[48]
KAUST-PI-2 ^b	1026	2368	490	98	2071	101	10.5	24.2	5.9	21.1	1	35	2.027	[48]
PIM1-CO15-50 ^b	770	1700	630	210	4600	370	3.7	8.1	4.6	21.9	1.8	30	0.507	[109]
PIM1-CO15-75 ^b	520	1100	350	110	2570	180	4.7	10	3	23.4	1.6	30	0.507	[109]
PIM-6FDA-OH	–	259	45.20	10.80	263	9.10	–	24	3.2	24.4	0.8	35	2.027	[110]
PIM-CO15 ^b	450	900	275	83	2000	130	5.4	10.8	4.2	24.1	1.6	30	0.507	[109]
PIM-CO19 ^b	880	2100	820	320	6100	580	2.8	6.6	3.3	19.1	1.8	30	0.507	[109]
PIMCO1-CO15-50 ^b	870	2100	760	240	5400	350	3.6	8.8	2.6	22.5	1.5	30	0.507	[109]
PIMCO2-CO15-50 ^b	950	2150	790	260	5300	430	3.7	8.3	3.2	20.4	1.7	30	0.507	[109]
PIMCO6-CO15-50 ^b	650	1500	520	170	3800	280	3.8	8.8	3	22.4	1.6	30	0.507	[109]
PIMCO19-CO15-50 ^b	620	1300	460	150	3400	260	4.1	8.7	3.1	22.7	1.7	30	0.507	[109]
PIM-EA-TB ^b	2570	7760	2150	525	7140	699	4.9	14.8	3.1	13.6	1.3	25	1.013	[49]
PIM-PI-1	260	530	150	47	1100	77	5.5	11.3	4.1	23.4	1.6	30	0.253	[45]
PIM-PI-2	160	220	39	9	210	9	17.8	24.4	3.2	23.3	1	30	0.253	[45]
PIM-PI-3	190	360	85	23	520	27	8.3	15.7	4.3	22.6	1.2	30	0.253	[45]
PIM-PI-4	205	300	64	16	420	20	12.8	18.8	3.7	26.3	1.3	30	0.253	[45]
PIM-PI-7	190	350	77	19	510	27	10	18.4	4	26.8	1.4	30	0.253	[45]
PIM-PI-8	425	1020	320	100	2270	170	4.3	10.2	4.1	22.7	1.7	30	0.253	[45]
PIM-PI-9	400	840	295	94	2180	170	4.3	8.9	3.2	23.2	1.8	25	1.013	[111]
PIM-PI-10	300	670	270	84	2154	168	3.6	8	3.1	25.6	2	25	1.013	[111]
PIM-PI-11	332	624	208	65	1523	129	5.1	9.6	3.2	23.4	2	25	1.013	[111]
PIM-PI-EA	1580	4230	1380	369	7340	457	4.3	11.5	3.2	19.9	1.2	25	1.013	[47]
PIM-PMDA-OH	–	190	6.90	30.50	198	7.70	–	6.2	3.7	6.5	0.3	35	2.027	[110]
PIM-SBF ^b	2200	6320	2640	786	13,900	1100	2.8	8	0.2	17.7	1.4	25	1.013	[34]
PIM-SBI-TB ^b	878	2200	720	232	2900	450	3.8	9.5	3.4	12.5	1.9	25	1.013	[49]

^a Large variation in experimentally measured values.^b Methanol conditioned membranes.**A.5. Diffusivity and diffusivity selectivity for selected ladder and semi-ladder polymers**

Polymer abbreviation	Diffusivity (10 ^{–8} cm ² /s)						Diffusivity selectivity					Temperature (°C)	Pressure (atm)	Reference
	He	H ₂	O ₂	N ₂	CO ₂	CH ₄	He/N ₂	H ₂ /N ₂	O ₂ /N ₂	CO ₂ /N ₂	CH ₄ /N ₂			
SBIPC	730	330	11	3	9	2	268.4	121.3	3.9	3.2	0.6	35	2	[21]
BBL	–	–	0.031	–	–	–	–	–	–	–	–	35	3	[24]
PBIBI-BPA	–	–	12	3	5	2	–	–	3.4	1.4	0.5	25	1	[108]
PBIBI-DOD	–	–	11	3	4	1	–	–	3.5	1.3	0.5	25	1	[108]
PBIBI-PPD	–	–	18	5	7	2	–	–	4	1.4	0.4	25	1	[108]

Appendix A (continued)

Polymer abbreviation	Diffusivity (10^{-8} cm ² /s)						Diffusivity selectivity					Temperature (°C)	Pressure (atm)	Reference
	He	H ₂	O ₂	N ₂	CO ₂	CH ₄	He/N ₂	H ₂ /N ₂	O ₂ /N ₂	CO ₂ /N ₂	CH ₄ /N ₂			
PBIBI-TDP	–	–	16	4	5	2	–	–	4.1	1.4	0.5	25	1	[108]
PIM-1 ^a	2700	1700	81	22	26	7	122.7	77.3	3.7	1.2	0.3	30	0.197	[25]
PIM-7	900	1100	62	16	21	5	56.3	68.8	3.9	1.3	0.3	30	0.197	[25]
Amine-PIM-1 ^b	5170	3970	223	60	39	18	86.5	66.4	3.7	0.7	0.3	25	1.014	[35]
AO-PIM-1 ^b	–	–	41	10	25	3	–	–	4.1	2.5	0.3	35	2.027	[36]
CARDO-PIM-1	1200	680	23	6	8	2	200	113.3	3.8	1.3	0.3	30	0.198	[27]
KAUST-PI-1 ^b	–	–	158	31	46	10	–	–	5.1	1.5	0.3	35	2.027	[48]
KAUST-PI-2 ^b	–	–	138	29	49	8	–	–	4.8	1.7	0.3	35	2.027	[48]
PIM1-CO15-50 ^b	4300	2800	170	65	68	26	66.2	43.1	2.6	1	0.4	30	0.507	[109]
PIM1-CO15-75 ^b	3300	2000	110	38	44	14	86.8	52.6	2.9	1.2	0.4	30	0.507	[109]
PIM-6FDA-OH	–	–	–	7	8	2	–	–	–	1.1	0.2	35	2.026	[110]
PIM-CO15 ^b	2600	1500	76	25	30	9	104.0	60	3.04	1.2	0.4	30	0.507	[109]
PIM-CO19 ^b	4100	2900	210	87	89	37	47.1	33.3	2.4	1	0.4	30	0.507	[109]
PIMCO1-CO15-50 ^b	4100	2900	170	55	66	19	74.5	52.7	3.1	1.2	0.3	30	0.507	[109]
PIMCO2-CO15-50 ^b	4400	3150	200	73	82	28	60.3	43.2	2.7	1.1	0.4	30	0.507	[109]
PIMCO6-CO15-50 ^b	3200	2200	120	43	51	16	74.4	51.2	2.8	1.2	0.4	30	0.507	[109]
PIMCO19-CO15-50 ^b	3300	2100	120	47	53	18	70.2	44.7	2.6	1.1	0.4	30	0.507	[109]
PIM-EA-TB ^b	10,000	7000	318	100	87	36	100	70	3.2	0.9	0.4	25	1.013	[49]
PIM-PI-9	–	–	163	56	64	22	–	–	2.9	1.1	0.4	25	1.013	[111]
PIM-PI-10	–	–	187	63	86	25	–	–	3	1.4	0.4	25	1.013	[111]
PIM-PI-EA	5070	3360	270	84	95	24	60.4	40	3.2	1.1	0.3	25	1.013	[47]
PIM-PMDA-OH	–	–	–	6	6	2	–	–	–	1	0.3	35	2.027	[110]
PIM-SBF ^b	7500	6800	420	129	181	42	58.1	52.7	3.3	1.4	0.3	25	1.013	[34]
PIM-SBI-TB ^b	5000	3500	201	75	74	32	66.7	46.7	2.7	1	0.4	25	1.013	[49]

^a Large variation in experimentally measured values.^b Methanol conditioned membranes.

A.6. Solubility and solubility selectivity for selected ladder and semi-ladder polymers

Polymer abbreviation	Solubility (cm ³ (STP)/cm ³ atm)						Solubility selectivity					Temperature (°C)	Pressure (atm)	Reference
	He	H ₂	O ₂	N ₂	CO ₂	CH ₄	He/N ₂	H ₂ /N ₂	O ₂ /N ₂	CO ₂ /N ₂	CH ₄ /N ₂			
SBI PC	0.05	0.13	0.50	0.38	2.60	0.93	0.1	0.3	1.3	6.8	2.4	35	2	[21]
BBL	–	–	0.52	–	–	–	–	–	–	–	–	35	3	[24]
PBIBI-BPA	–	–	1.05	0.64	13	1	–	–	1.6	20.4	1.6	25	1	[108]
PBIBI-DOD	–	–	1.03	0.62	12.80	0.90	–	–	1.7	20.8	1.5	25	1	[108]
PBIBI-PPD	–	–	1.09	0.81	15.70	1.54	–	–	1.3	19.3	1.9	25	1	[108]
PBIBI-TDP	–	–	0.96	0.63	13.10	1.03	–	–	1.5	20.8	1.6	25	1	[108]
PIM-1 ^a	0.18	0.58	3.50	3.19	66.87	13.68	0.1	0.2	1.1	21	4.3	30	0.197	[25]
PIM-7	0.38	0.61	2.36	1.98	39.51	9.12	0.2	0.3	1.2	20	4.6	30	0.197	[25]
Amine-PIM-1 ^b	0.18	0.59	3.06	2.92	36.58	12.87	0.1	0.2	1	12.5	4.4	25	1.014	[35]
AO-PIM-1 ^b	–	–	2.76	2.52	35.56	9.88	–	–	1.1	14.1	3.9	35	2.027	[36]
CARDO-PIM-1	0.12	0.38	1.90	1.60	42.60	8.36	0.1	0.2	1.2	26.6	5.2	30	0.198	[27]
KAUST-PI-1 ^b	–	–	3.02	2.61	40.50	8.51	–	–	1.2	15.5	3.3	35	2.027	[48]
KAUST-PI-2 ^b	–	–	2.71	2.58	31.91	9.57	–	–	1.1	12.4	3.7	35	2.027	[48]
PIM1-CO15-50 ^b	0.14	0.46	2.82	2.45	51.40	10.81	0.1	0.2	1.1	20.9	4.4	30	0.507	[109]
PIM1-CO15-75 ^b	0.12	0.42	2.42	2.20	44.38	9.77	0.1	0.2	1.1	20.2	4.4	30	0.507	[109]
PIM-6FDA-OH	–	–	–	1.13	24.90	4.19	–	–	–	22	3.7	35	2.027	[110]
PIM-CO15 ^b	0.13	0.46	2.75	2.52	50.66	10.86	0.1	0.2	1.1	20.1	4.3	30	0.507	[109]
PIM-CO19 ^b	0.16	0.55	2.97	2.79	52.08	11.91	0.1	0.2	1.1	18.6	4.3	30	0.507	[109]
PIMCO1-CO15-50 ^b	0.16	0.55	3.40	3.32	62.17	14	0.05	0.2	1	18.8	4.2	30	0.507	[109]

(continued on next page)

Appendix A (continued)

Polymer abbreviation	Solubility (cm ³ (STP)/cm ³ atm)						Solubility selectivity					Temperature (°C)	Pressure (atm)	Reference
	He	H ₂	O ₂	N ₂	CO ₂	CH ₄	He/ N ₂	H ₂ / N ₂	O ₂ / N ₂	CO ₂ / N ₂	CH ₄ / N ₂			
PIMCO2-CO15-50 ^b	0.16	0.52	3	2.71	49.11	11.67	0.1	0.2	1.1	18.1	4.3	30	0.507	[109]
PIMCO6-CO15-50 ^b	0.15	0.52	3.29	3	56.62	13.30	0.05	0.2	1.1	18.9	4.4	30	0.507	[109]
PIMCO19-CO15-50 ^b	0.14	0.47	2.91	2.43	48.75	10.98	0.1	0.2	1.2	20.1	4.5	30	0.507	[109]
PIM-EA-TB ^b	0.20	0.81	6.08	4.76	57.76	15	0.04	0.2	1.3	12.1	3.1	25	1.013	[49]
PIM-PI-9	–	–	2.97	2.79	38.30	11.35	–	–	1.1	13.7	4.1	25	1.013	[111]
PIM-PI-10	–	–	2.87	2.74	35.87	10.74	–	–	1	13.1	3.9	25	1.013	[111]
PIM-PI-EA	0.23	0.95	3.88	3.33	58.57	14.39	0.1	0.3	1.2	17.6	4.3	25	1.013	[47]
PIM-PMDA-OH	–	–	–	0.95	26.30	3.92	–	–	–	27.7	4.1	35	2.027	[110]
PIM-SBF ^b	0.20	0.71	4.76	4.61	53.91	19.86	0.04	0.2	1	11.7	4.3	25	1.013	[34]
PIM-SBI-TB ^b	0.12	0.48	2.74	2.33	29.79	10.84	0.1	0.2	1.2	12.8	4.7	25	1.013	[49]

^a Large variation in experimentally measured values.^b Methanol conditioned membranes.**A.7. Permeability and permselectivity for selected poly(ionic liquid)s**

Polymer abbreviation	Permeability (Barrer)				Permselectivity			Temperature (°C)	Pressure (atm)	Reference
	H ₂	N ₂	CO ₂	CH ₄	H ₂ /CO ₂	CO ₂ /N ₂	CO ₂ /CH ₄			
P[GC ₆ IM]	4.80	0.20	4.40	0.16	1.1	22	27.5	20	2	[75]
P[GPEG ₁ IM]	4.60	0.14	4	0.12	1.2	28.6	33.3	20	2	[75]
P[GPEG ₂ IM]	4	0.16	3.80	0.13	1.1	23.8	29.2	20	2	[75]
P[IOIM][Br]	0.84	–	0.13	–	6.5	–	–	22	2	[76]
P[IOIM]-[C ₆ mim][Tf ₂ N]	–	0.83	19	1.10	–	22.9	17.3	22	2	[76]
P[IOIM][Tf ₂ N]	5.30	0.22	5.30	0.26	1	24.1	20.4	22	2	[76]
P[MAC ₄ IM][Tf ₂ N]	–	0.71	22	0.97	–	31	22.7	20	2	[73]
P[MAMIM][Tf ₂ N]	–	0.23	7	0.19	–	30.4	36.8	20	2	[73]
P[VBC ₃ CNIM][Tf ₂ N]	–	0.11	4.10	0.11	–	37.3	37.3	22	2	[74]
P[VBC ₅ CNIM][Tf ₂ N]	–	0.21	8.20	0.28	–	39	29.3	22	2	[74]
P[VBIM][dca]	–	0.16	4.24	–	–	26.5	–	35	1	[112]
P[VBIM][dca]-[C ₂ mim][B(CN) ₄] (1:1)	–	4.67	184.70	–	–	39.6	–	35	1	[112]
P[VBIM][dca]-[C ₂ mim][BF ₄] (1:1)	–	1.79	74.80	–	–	41.8	–	35	1	[112]
P[VBIM][dca]-[C ₂ mim][dca] (1:1)	–	2.40	118.80	–	–	49.5	–	35	1	[112]
P[VBMIM]-[C ₂ mim][Tf ₂ N]	–	1.10	44	1.60	–	40	27.5	22	2	[77]
P[VBMIM][Tf ₂ N]	–	0.29	9.20	0.24	–	31.7	38.3	22	2	[77]
P[VBOEG ₁ IM][Tf ₂ N]	–	0.39	16	0.48	–	41	33.3	22	2	[74]
P[VBOEG ₂ IM][Tf ₂ N]	–	0.50	22	0.74	–	44	29.7	22	2	[74]
P[VBOEG ₁]-[C ₂ mim][dca]	–	1.10	41	1.60	–	37.3	25.6	22	2	[68]
P[VBOEG ₁]-[C ₂ mim][OTf]	–	1.20	43	1.70	–	35.8	25.3	22	2	[68]
P[VBOEG ₁]-[C ₂ mim][SbF ₆]	–	1.10	42	1.50	–	38.2	28	22	2	[68]
P[VBOEG ₁]-[C ₂ mim][Tf ₂ N]	–	1.70	60	2.60	–	35.3	23.1	22	2	[68]
P[VBOEG ₁]-[C ₄ FAmim][Tf ₂ N]	–	1.31	47	1.74	–	36	1.3	22	2	[78]
P[VBOEG ₁]-[C ₄ mim][Tf ₂ N]	–	1.38	51	2.13	–	37	1.5	22	2	[78]
P[VBOEG ₁]-[C ₃ CNmim][Tf ₂ N]	–	0.83	33	1.18	–	40	1.4	22	2	[78]
P[VBOEG ₁]-[OEGmim][Tf ₂ N]	–	1.39	53	2.04	–	38	1.5	22	2	[78]
P[VBOEG ₁]-[Smim][Tf ₂ N]	–	1.67	55	2.75	–	33	1.7	22	2	[78]
P[VBOEG ₁][Tf ₂ N]	–	0.39	16	0.48	–	41	1.2	22	2	[78]
P[VMMIM][Tf ₂ N]	–	0.29	9.2	0.24	–	31.7	38.3	20	2	[73]
P[VC ₄ MIM][Tf ₂ N]	–	0.67	20	0.91	–	29.9	22	20	2	[73]
P[VC ₆ MIM][Tf ₂ N]	–	1.40	32	2.30	–	22.9	13.9	20	2	[73]
P[VEIM][dca]-[C ₂ mim][B(CN) ₄] (1:1)	–	2.23	99.80	–	–	44.8	–	35	1	[112]
P[VEIM][dca]-[C ₂ mim][BF ₄] (1:1)	–	0.79	36	–	–	45.6	–	35	1	[112]
P[VEIM][dca]-[C ₂ mim][dca] (1:1)	–	1.03	62.10	–	–	60.3	–	35	1	[112]
P[VHIM][dca]	–	1.60	33.50	–	–	20.9	–	35	1	[112]
P[VHIM][dca]-[C ₂ mim][B(CN) ₄] (1:1)	–	6.42	230	–	–	35.8	–	35	1	[112]
P[VHIM][dca]-[C ₂ mim][BF ₄] (1:1)	–	2.35	89.70	–	–	38.2	–	35	1	[112]
P[VHIM][dca]-[C ₂ mim][dca] (1:1)	–	3.28	138	–	–	42.1	–	35	1	[112]

A.8. Diffusivity and diffusivity selectivity for selected poly(ionic liquid)s

Polymer abbreviation	Diffusivity (10 ⁻⁸ cm ² /s)			Diffusivity selectivity		Temperature (°C)	Pressure (atm)	Reference
	N ₂	CO ₂	CH ₄	CO ₂ /N ₂	CO ₂ /CH ₄			
P[GC ₆ IM]	–	1.20	–	–	–	20	2	[75]
P[GPEG ₁ IM]	–	0.78	–	–	–	20	2	[75]
P[GPEG ₂ IM]	–	0.83	–	–	–	20	2	[75]
P[IOIM][Br]	–	0.23	–	–	–	22	2	[76]
P[IOIM]–[C ₆ mim][Tf ₂ N]	–	12	–	–	–	22	2	[76]
P[IOIM][Tf ₂ N]	–	5	–	–	–	22	2	[76]
P[MAC ₄ IM][Tf ₂ N]	–	3.60	1.27	–	2.8	20	2	[73]
P[MAMIM][Tf ₂ N]	–	1.50	0.89	–	1.7	20	2	[73]
P[VBIM][dca]	4	4.42	–	1.1	–	35	1	[112]
P[VBIM][dca]–[C ₂ mim][B(CN) ₄] (1:1)	69.70	78.20	–	1.1	–	35	1	[112]
P[VBIM][dca]–[C ₂ mim][BF ₄] (1:1)	42.62	46.75	–	1.1	–	35	1	[112]
P[VBIM][dca]–[C ₂ mim][dca] (1:1)	70.59	77.44	–	1.1	–	35	1	[112]
P[VBOEGI]–[C ₄ mim][Tf ₂ N]	–	9.50	–	–	–	22	2	[78]
P[VBOEGI]–[C ₄ Famim][Tf ₂ N]	–	8.10	–	–	–	22	2	[78]
P[VBOEGI]–[C ₃ CNmim][Tf ₂ N]	–	5.20	–	–	–	22	2	[78]
P[VBOEGI]–[OEGmim][Tf ₂ N]	–	10.80	–	–	–	22	2	[78]
P[VBOEGI]–[Smim][Tf ₂ N]	–	10.60	–	–	–	22	2	[78]
P[VBOEGI][Tf ₂ N]	–	3.60	–	–	–	22	2	[78]
P[VMMIM][Tf ₂ N]	–	1.70	0.88	–	1.9	20	2	[73]
P[VC ₄ MIM][Tf ₂ N]	–	3.50	1.28	–	2.7	20	2	[73]
P[VC ₆ MIM][Tf ₂ N]	11	7.70	3.10	0.7	2.5	20	2	[73]
P[VEIM][dca]–[C ₂ mim][B(CN) ₄] (1:1)	36.56	46.14	–	1.3	–	35	1	[112]
P[VEIM][dca]–[C ₂ mim][BF ₄] (1:1)	22.57	25.99	–	1.2	–	35	1	[112]
P[VEIM][dca]–[C ₂ mim][dca] (1:1)	36.79	45	–	1.2	–	35	1	[112]
P[VHIM][dca]	25	23.53	–	0.9	–	35	1	[112]
P[VHIM][dca]–[C ₂ mim][B(CN) ₄] (1:1)	83.28	87.79	–	1.1	–	35	1	[112]
P[VHIM][dca]–[C ₂ mim][BF ₄] (1:1)	47.78	51.61	–	1.1	–	35	1	[112]
P[VHIM][dca]–[C ₂ mim][dca] (1:1)	76.28	81.86	–	1.1	–	35	1	[112]

A.9. Solubility and solubility selectivity for selected poly(ionic liquid)s

Polymer abbreviation	Solubility (cm ³ (STP)/cm ³ atm)			Solubility selectivity		Temperature (°C)	Pressure (atm)	Reference
	N ₂	CO ₂	CH ₄	CO ₂ /N ₂	CO ₂ /CH ₄			
P[GC ₆ IM]	–	2.80	–	–	–	20	2	[75]
P[GPEG ₁ IM]	–	3.80	–	–	–	20	2	[75]
P[GPEG ₂ IM]	–	3.50	–	–	–	20	2	[75]
P[IOIM][Br]	–	0.45	–	–	–	22	2	[76]
P[IOIM]–[C ₆ mim][Tf ₂ N]	–	1.20	–	–	–	22	2	[76]
P[IOIM][Tf ₂ N]	–	0.79	–	–	–	22	2	[76]
P[MAC ₄ IM][Tf ₂ N]	–	4.50	0.59	–	7.7	20	2	[73]
P[MAMIM][Tf ₂ N]	–	3.60	0.17	–	21	20	2	[73]
P[VBIM][dca]	0.03	0.73	–	24	–	35	1	[112]
P[VBIM][dca]–[C ₂ mim][B(CN) ₄] (1:1)	0.05	1.80	–	35.2	–	35	1	[112]
P[VBIM][dca]–[C ₂ mim][BF ₄] (1:1)	0.03	1.21	–	38	–	35	1	[112]
P[VBIM][dca]–[C ₂ mim][dca] (1:1)	0.03	1.17	–	45.9	–	35	1	[112]
P[VBOEGI]–[C ₄ mim][Tf ₂ N]	–	4.09	–	–	–	22	2	[78]
P[VBOEGI]–[C ₄ Famim][Tf ₂ N]	–	4.39	–	–	–	22	2	[78]
P[VBOEGI]–[C ₃ CNmim][Tf ₂ N]	–	4.85	–	–	–	22	2	[78]
P[VBOEGI]–[OEGmim][Tf ₂ N]	–	3.71	–	–	–	22	2	[78]
P[VBOEGI]–[Smim][Tf ₂ N]	–	3.94	–	–	–	22	2	[78]
P[VBOEGI][Tf ₂ N]	–	3.37	–	–	–	22	2	[78]
P[VMMIM][Tf ₂ N]	–	4.00	0.21	–	–	20	2	[73]

(continued on next page)

Appendix A (continued)

Polymer abbreviation	Solubility (cm ³ (STP)/cm ³ atm)			Solubility selectivity		Temperature (°C)	Pressure (atm)	Reference
	N ₂	CO ₂	CH ₄	CO ₂ /N ₂	CO ₂ /CH ₄			
P[VC ₄ MIM][Tf ₂ N]	–	4.40	0.55	–	8	20	2	[73]
P[VC ₆ MIM][Tf ₂ N]	0.10	3.90	0.57	39	6.9	20	2	[73]
P[VEIM][dca]-[C ₂ mim][B(CN) ₄] (1:1)	0.05	1.67	–	36	–	35	1	[112]
P[VEIM][dca]-[C ₂ mim][BF ₄] (1:1)	0.03	1.05	–	40	–	35	1	[112]
P[VEIM][dca]-[C ₂ mim][dca] (1:1)	0.02	1.05	–	50.9	–	35	1	[112]
P[VHIM][dca]	0.05	1.08	–	22.2	–	35	1	[112]
P[VHIM][dca]-[C ₂ mim][B(CN) ₄] (1:1)	0.06	1.99	–	34	–	35	1	[112]
P[VHIM][dca]-[C ₂ mim][BF ₄] (1:1)	0.04	1.32	–	35.4	–	35	1	[112]
P[VHIM][dca]-[C ₂ mim][dca] (1:1)	0.03	1.36	–	41.5	–	35	1	[112]

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